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PHYSICS • UPPER SIXTH | UPPER SIXTH — PHYSICS

CHAPTER 1

GRAVITATIONAL FIELDS

SUGGESTED TIME: 4 PERIODS

IN THIS CHAPTER

This chapter studies gravitational fields: Newton's inverse-square law, Kepler's laws, the variation of g, gravitational potential and potential energy, escape and orbital speeds, geostationary satellites, and projectile motion in a uniform field. It provides the essential tools for the GCE Advanced Level mechanics paper.

- By the end of this chapter I can state and apply Newton's law of universal gravitation between two point masses
- By the end of this chapter I can state Kepler's three laws and use the third law to relate orbital period and radius
- By the end of this chapter I can define gravitational field strength and derive $g = GM/r^2$ for a planet
- By the end of this chapter I can describe and calculate the variation of g with altitude, depth and latitude
- By the end of this chapter I can define gravitational potential and potential energy and compute the work done moving a mass in a field
- By the end of this chapter I can derive and calculate escape velocity and orbital speed for a satellite

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THIS SIGNED COURSE CONTAINS

COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Newton's Law of Gravitation and Kepler's Laws
2. Gravitational Field Strength and Variation of g
3. Gravitational Potential, Potential Energy, Escape and Orbital Speeds
4. Motion in the Gravitational Field: Projectile Motion
5. Integration activity
6. Methods and worked examples
7. Exercises
8. Solutions to the exercises
9. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can state and apply Newton's law of universal gravitation between two point masses
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- By the end of this chapter I can derive and calculate escape velocity and orbital speed for a satellite
- By the end of this chapter I can explain the conditions for a geostationary orbit and compute its radius
- By the end of this chapter I can analyse projectile motion in a uniform gravitational field (range, maximum height, time of flight)

PREREQUISITES

- Newton's laws of motion and the concept of weight (Lower Sixth mechanics)

- Uniform circular motion: centripetal acceleration $a = v^2/r = \omega^2 r$ (Lower Sixth)
- Work, kinetic energy and potential energy in a uniform field (Lower Sixth)
- Projectile motion basics from Lower Sixth mechanics
- Algebraic manipulation of powers, square roots and scientific notation

1 NEWTON'S LAW OF GRAVITATION AND KEPLER'S LAWS

Every evening, the CRTV signal reaches your television from a satellite parked 35 786 km above the equator — yet nobody ever launched it with a ladder or a rope. How fast must a satellite move to "hang" above the same point of Cameroon without falling? Why does a mango dropped in Bamenda fall slightly differently from one dropped at the top of Mount Cameroon? And why does a stone thrown horizontally from a school field in Buea follow a curved path? This chapter answers these questions with one single idea: **Newton's law of universal gravitation**.

1.1 From falling bodies to planetary motion: a historical sketch

For centuries, people believed that the physics of the heavens was different from the physics of the Earth. A mango falls from a tree in Bamenda; the Moon circles the Earth; the planets wander across the sky. These looked like three unrelated phenomena.

The story changed in the seventeenth century. First, **Galileo** showed that all bodies near the Earth's surface fall with the same acceleration g (air resistance apart). Then **Tycho Brahe** accumulated decades of precise observations of planetary positions, and his assistant **Johannes Kepler** extracted from this data three empirical laws describing how planets move around the Sun.

The genius of **Isaac Newton** (around 1687, in his Principia) was to realise that one single force explains all of it. The famous — possibly legendary — apple suggested the question: if gravity pulls the apple down, does the same pull reach the Moon? Newton's answer: yes. The Moon is continuously "falling" towards the Earth, but its sideways velocity is so large that it perpetually misses. The same attractive force that drops a mango also steers the Moon and the planets. This is why the law is called universal.

DID YOU KNOW? **A force you use every day without knowing it**

Every GPS position, every satellite TV channel, every tide table on the coast of Kribi and Limbe is computed from Newton's law of gravitation, written down more than three centuries ago. The same equation that describes a falling mango describes a satellite 35 786 km above the Atlantic.

1.2 Newton's law of universal gravitation

DEFINITION **Newton's law of universal gravitation**

Every particle of matter in the universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them:

$$F = \frac{G\,M\,m}{r^2}$$

where F is the magnitude of the gravitational force (in N), M and m are the two masses (in

kg), r is the distance between their centres (in m), and G is the **universal gravitational constant**:

$$G = 6.67 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}.$$

Because the force varies as $1/r^2$, we say that gravitation obeys an **inverse-square law**: doubling the separation divides the force by four, tripling it divides the force by nine. This $1/r^2$ dependence is the single most important feature of the law, and it reappears everywhere in this chapter (in g, in the potential, in orbital mechanics).

The law is stated for **point masses**. Newton proved (using his newly invented calculus) that a uniform spherical body attracts external objects exactly as if all its mass were concentrated at its centre. This is why we may treat the Earth, the Sun and the planets as point masses located at their centres.

WARNING **Distance measured between centres, not surfaces**

In $F = GMm/r^2$, the distance r is measured between the centres of the two bodies, never between their surfaces. For a satellite at altitude h above the Earth, $r = R_E + h$, where $R_E = 6.37 \times 10^6$ m is the Earth's radius. Forgetting to add R_E is one of the most frequent errors at the GCE Advanced Level.



Two point masses M and m separated by r. The gravitational forces form an action–reaction pair: equal in magnitude, opposite in direction, always attractive.

Vector nature and superposition.

Gravitation is a vector force. On mass m at position $\vec{r}$ relative to mass M,

$$\vec{F} = -\frac{GMm}{r^2}\,\hat{r},$$

where the minus sign and the unit vector $\hat{r}$ (pointing from M towards m) express that the force is always **attractive**, directed along the line joining the centres. The two forces on M and on m form a Newton's-third-law pair: equal magnitudes, opposite directions. When several masses are present, the total force on one body is the vector sum of the individual forces (principle of **superposition**). For example, the force on the Moon is the vector sum of the pulls of the Earth and the Sun.

The constant G and its measurement.

G is a universal constant: it has the same value everywhere in the universe, for all pairs of masses.

Its unit follows directly from the law: $[G] = [F][r^2]/[m^2]$, that is $\mathrm{N\,m^2\,kg^{-2}}$. Its smallness ($\sim 10^{-11}$) explains why gravitational attraction between everyday objects is utterly negligible compared with the pull of the whole Earth. G was first measured by **Cavendish** (1798) with a torsion balance: two small lead spheres are hung from a fine fibre, two large lead spheres are brought near them, and the tiny twist of the fibre — calibrated against a known restoring torque — gives the attractive force, hence G.

WORKED EXAMPLE 1 **Everyday masses versus astronomical masses**

Statement. (a) Calculate the gravitational force between two students of masses 60 kg and 55 kg standing 1.0 m apart. (b) Calculate the force between the Earth ($M_E = 5.97 \times 10^{24}$ kg) and the same 60 kg student standing on its surface ($R_E = 6.37 \times 10^6$ m). Comment.

SOLUTION

Solution. (a) Treating the students as point masses:

$$F = \frac{Gm_1m_2}{r^2} = \frac{(6.67 \times 10^{-11})(60)(55)}{(1.0)^2} = 2.2 \times 10^{-7}\ \mathrm{N}.$$

This is about the weight of a grain of sand — impossible to feel. (b) The student is at distance $r = R_E$ from the Earth's centre:

$$F = \frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(60)}{(6.37 \times 10^6)^2} = \frac{2.39 \times 10^{16}}{4.06 \times 10^{13}} = 5.9 \times 10^2\ \mathrm{N}.$$

This is simply the student's weight, $mg \approx 60 \times 9.81 = 5.9 \times 10^2$ N. The Earth's enormous mass compensates the smallness of G.

1.3 Weight and the link between g, G, M_E and R_E

The **weight** of a body of mass m at the Earth's surface is nothing other than the gravitational force exerted on it by the whole Earth:

$$mg = \frac{GM_Em}{R_E^2} \quad \Longrightarrow \quad \boxed{g = \frac{GM_E}{R_E^2}}.$$

This relation is extremely useful: knowing g, G and R_E, Cavendish's experiment effectively allowed scientists to "weigh the Earth", since $M_E = gR_E^2/G$. Numerically, $g = (6.67 \times 10^{-11})(5.97 \times 10^{24})/(6.37 \times 10^6)^2 \approx 9.81\ \mathrm{m\,s^{-2}}$, as expected.

KEY POINTS **G is universal, g is local**

$G = 6.67 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$ has the same value everywhere in the universe. The gravitational field strength g is local: it depends on the mass and radius of the attracting body and on the position. At the Earth's surface $g \approx 9.81\ \mathrm{m\,s^{-2}}$; on the Moon $g \approx 1.6\ \mathrm{m\,s^{-2}}$; at altitude, g

decreases. Never confuse the two symbols in an examination.

1.4 Kepler's laws of planetary motion

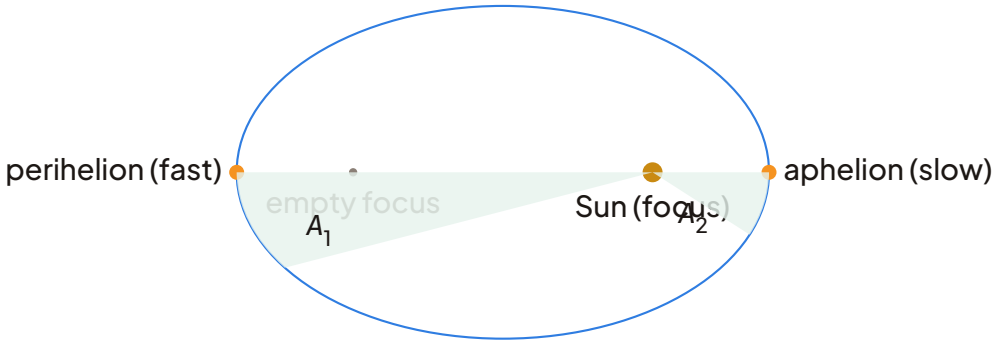
Before Newton, Kepler had summarised Brahe's observations in three laws. Newton later showed that all three follow from $F = GMm/r^2$ — a triumph of the universal law.

PROPERTY **Kepler's three laws**

1. Law of orbits: each planet moves on an **ellipse** with the Sun at one focus.
2. Law of areas: the line joining the Sun to a planet sweeps out equal areas in equal times. Consequently a planet moves faster near the Sun (perihelion) and slower far from it (aphelion).
3. Law of periods: the square of the orbital period is proportional to the cube of the orbit's semi-major axis:

$$T^2 = \frac{4\pi^2}{GM_S} r^3,$$

where M_S is the mass of the central body (the Sun for planets, the Earth for satellites).



Kepler's first and second laws: the orbit is an ellipse with the Sun at one focus; the radius vector sweeps equal areas in equal times, so the planet travels faster near perihelion.

METHOD **Deriving Kepler's third law for a circular orbit (examinable)**

For a circular orbit of radius r around a central mass M:

1. The gravitational force provides the centripetal force:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}.$$

2. Simplify: $v^2 = \frac{GM}{r}$, so the orbital speed is $v = \sqrt{GM/r}$. Note that the satellite's mass m cancels: the orbit depends only on the central mass.

3. Write the speed in terms of the period: $v = \frac{2\pi r}{T}$.

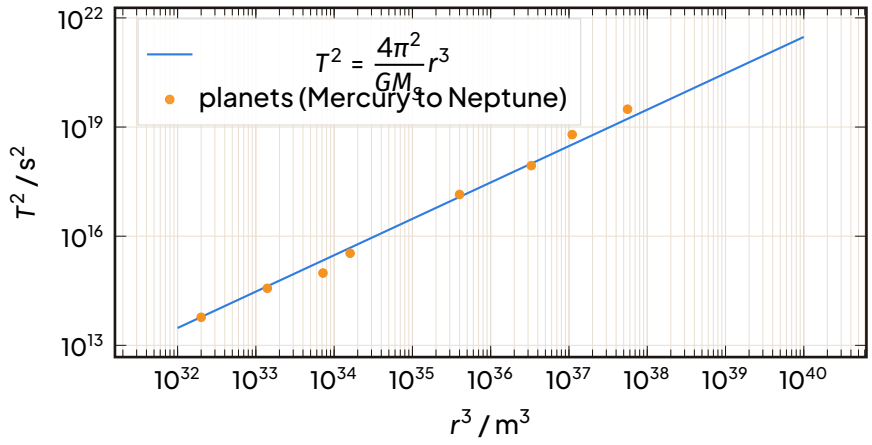
4. Substitute and square:

$$\frac{4\pi^2 r^2}{T^2} = \frac{GM}{r} \quad \Longrightarrow \quad T^2 = \frac{4\pi^2}{GM} r^3.$$

Hence $T^2/r^3 = 4\pi^2/(GM)$ is the same constant for all bodies orbiting the same central mass.

WARNING **SI units in Kepler's third law**

Before substituting in $T^2 = 4\pi^2 r^3/(GM)$, convert r to metres and expect T in seconds. Only convert to minutes, hours or days after the calculation. Mixing kilometres with G in SI units is a classic source of wrong answers.



Log–log plot of T^2 against r^3 for the planets of the solar system. All points lie on a straight line of slope $4\pi^2/(GM_S)$, confirming Kepler's third law.

WORKED EXAMPLE 2 **Period of the Moon from Kepler's third law**

Statement. The Moon orbits the Earth at a mean distance $r = 3.84 \times 10^8$ m from the Earth's centre. Taking $M_E = 5.97 \times 10^{24}$ kg, calculate its orbital period in days.

SOLUTION

Solution. Apply Kepler's third law with the Earth as central body:

$$T^2 = \frac{4\pi^2 r^3}{GM_E} = \frac{4\pi^2 (3.84 \times 10^8)^3}{(6.67 \times 10^{-11})(5.97 \times 10^{24})}$$

$$= \frac{39.48 \times 5.67 \times 10^{25}}{3.98 \times 10^{14}} = 5.62 \times 10^{12}\ \text{s}^2.$$

$$T = 2.37 \times 10^6\ \text{s} = \frac{2.37 \times 10^6}{86\,400}\ \text{days} \approx 27.4\ \text{days}.$$

This matches the known sidereal period of the Moon — a beautiful confirmation of the law.

EXERCISE 1 **Force and period**

1. Calculate the gravitational force between the Sun ($M_S = 1.99 \times 10^{30}$ kg) and the Earth ($M_E = 5.97 \times 10^{24}$ kg), separated by $r = 1.50 \times 10^{11}$ m.
2. A satellite orbits the Earth at $r = 7.0 \times 10^6$ m from the centre. Using Kepler's third law, find its period in minutes.

SOLUTION **Exercise 1**

1. $F = \dfrac{(6.67 \times 10^{-11})(1.99 \times 10^{30})(5.97 \times 10^{24})}{(1.50 \times 10^{11})^2} = \dfrac{7.93 \times 10^{44}}{2.25 \times 10^{22}} = 3.52 \times 10^{22}\ \text{N}.$

2. $T^2 = \dfrac{4\pi^2 (7.0 \times 10^6)^3}{(6.67 \times 10^{-11})(5.97 \times 10^{24})} = \dfrac{1.354 \times 10^{22}}{3.98 \times 10^{14}} = 3.40 \times 10^7\ \text{s}^2$, so $T = 5.83 \times 10^3\ \text{s} \approx 97\ \text{min}.$

KEY POINTS **Key points of this section**

- Newton's law: $F = GMm/r^2$, attractive, along the line of centres; r is measured centre to centre. The force obeys an inverse-square law.
- G is universal; $g = GM_E/R_E^2$ is local.

- Kepler: elliptical orbits (Sun at a focus); equal areas in equal times; $T^2 = 4\pi^2 r^3/(GM)$, derived by equating gravitation to centripetal force.

2 GRAVITATIONAL FIELD STRENGTH AND VARIATION OF g

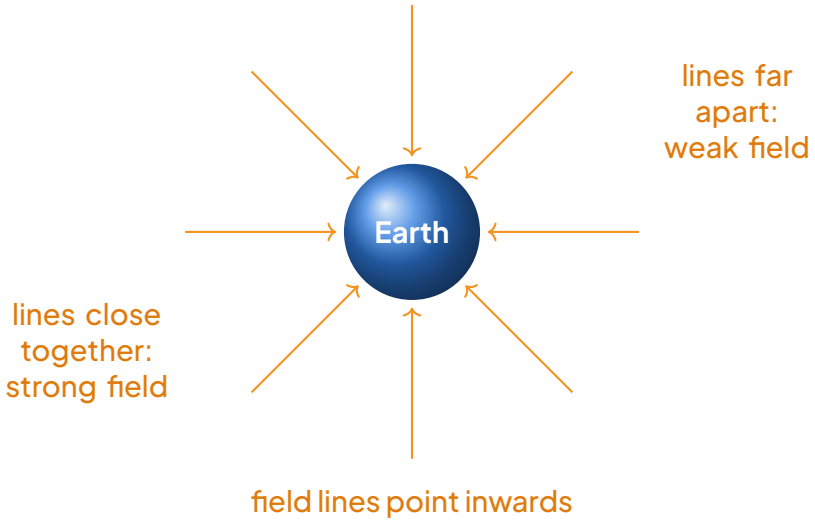
In the previous section we established Newton's law of universal gravitation: any two masses attract each other with a force $F = GMm/r^2$, and we saw how this single law explains Kepler's empirical laws of planetary motion. We now change our point of view. Instead of always speaking of a pair of masses, we focus on one mass — the Earth — and describe the "influence" it creates in the whole space around it. This influence is what physicists call a **gravitational field**. The idea is powerful: once we know the field, we can predict the force on any object placed at any point, without referring back to the Earth each time.

2.1 Qualitative Description of the Earth's Gravitational Field

Drop a stone anywhere near the Earth's surface: it falls towards the ground. Release a stone above Buea, above Douala, or above the ocean: in every case it accelerates towards the centre of the Earth. The Moon, far away, is also continuously "falling" towards the Earth, which is what keeps it in orbit. These observations show that:

- the Earth exerts an attractive influence on every mass around it, near or far;
- this influence is directed towards the centre of the Earth at every point;
- the influence becomes weaker as we move further away.

We picture this influence using **field lines**. Because the Earth is (to a good approximation) spherical, the field lines are **radial**, like the spokes of a wheel, and they are always directed **inwards**, towards the centre, because gravity is always attractive. Where the lines are close together (near the surface), the field is strong; where they spread out (far away), the field is weak.



Radial gravitational field lines around the Earth. Every arrow points towards the centre; the density of lines decreases with distance, showing that the field weakens as $1/r^2$.

WARNING **Field lines are a picture, not a physical object**

Nothing material exists along a field line. Field lines are a representation invented to visualise the direction and relative strength of the field. The field itself exists at every point of space, whether or not a line is drawn through it.

2.2 Definition of Gravitational Field Strength

Suppose we want to compare the strength of gravity at two different places, say at sea level and at the summit of Mount Cameroon. We cannot simply compare the forces on two objects, because the force depends on the object's mass: a heavy lorry is pulled harder than a small stone, even at the same place. To characterise the field itself, we must remove the mass of the test object from the comparison. This leads to the following definition.

DEFINITION **Gravitational field strength**

The **gravitational field strength** $\vec{g}$ at a point is the gravitational force per unit mass experienced by a small test mass m placed at that point:

$$\vec{g} = \frac{\vec{F}}{m} \qquad \text{or in magnitude} \qquad g = \frac{F}{m}.$$

Its SI unit is the newton per kilogram, $1\,\mathrm{N\,kg^{-1}}$. Since $1\,\mathrm{N} = 1\,\mathrm{kg\,m\,s^{-2}}$, we have the equivalent unit

$$1\,\mathrm{N\,kg^{-1}} = 1\,\mathrm{m\,s^{-2}},$$

which shows that g is none other than the acceleration of free fall at that point. The vector $\vec{g}$ always points in the direction of the force, i.e. towards the centre of the attracting mass.

KEY POINTS

g is a property of the point in space, not of the object placed there. A feather and a hammer at the same point experience the same g (and, in vacuum, fall with the same acceleration), even though the forces on them are very different.

2.3 Field Strength Due to a Point or Spherical Mass

Consider a point mass M. Place a test mass m at a distance r from it. By Newton's law of gravitation, the force on m is

$$F = \frac{GMm}{r^2}.$$

Applying the definition $g = F/m$, the mass m cancels:

PROPERTY **Field of a point or spherical mass**

The gravitational field strength at distance r from a point mass M — or outside a uniform spherical mass M, at distance r from its centre — is

$$g = \frac{GM}{r^2}.$$

The field is radial and directed towards M. (The proof for a sphere requires integration over the whole sphere and is not examinable; the result, due to Newton, is that a uniform sphere attracts external bodies exactly as if all its mass were concentrated at its centre.)

Applying this to the Earth, of mass M_E and radius R_E, the field strength at its surface is

$$g_0 = \frac{GM_E}{R_E^2}.$$

EXEMPLE **The Earth's surface field**

With $G = 6.67e-11\,\mathrm{N\,m^2\,kg^{-2}}$, $M_E = 5.98e24\,\mathrm{kg}$ and $R_E = 6.37e6\,\mathrm{m}$:

$$g_0 = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{(6.37 \times 10^{6})^2} \approx 9.83\,\mathrm{N\,kg^{-1}}.$$

This is the familiar acceleration of free fall, $g_0 \approx 9.81\,\mathrm{m\,s^{-2}}$.

2.4 The Uniform Field Approximation Near the Surface

Since $g = GM_E/r^2$ changes with r, the field is strictly not uniform. However, near the surface the change is tiny: climbing from sea level to the top of Mount Cameroon ($h \approx 4.1\,\text{km}$) changes r from 6371 km to 6375 km, a relative change of less than 0.1%. Over such small heights we may treat $\vec{g}$ as **constant in magnitude and direction**: the field is approximately **uniform**, with parallel, equally spaced field lines pointing "downwards". This is exactly the approximation used in Lower Sixth projectile motion, and we shall use it again in Section 4.

2.5 Variation of g with Altitude (Outside the Earth)

At altitude h above the surface, the distance from the centre is $r = R_E + h$. Hence

$$g(h) = \frac{GM_E}{(R_E + h)^2}.$$

Dividing by $g_0 = GM_E/R_E^2$ eliminates G and M_E:

PROPERTY **Variation of g with altitude**

$$\boxed{g(h) = g_0 \frac{R_E^2}{(R_E + h)^2}}$$

For small altitudes ($h \ll R_E$), the binomial approximation $(1 + x)^{-2} \approx 1 - 2x$ gives

$$g(h) \approx g_0\left(1 - \frac{2h}{R_E}\right).$$

METHOD **Computing g at altitude using ratios**

1. Write the ratio $\frac{g}{g_0} = \frac{R_E^2}{(R_E + h)^2}$ — no need for G or M_E.
2. Convert h and R_E to the same unit (usually metres).
3. Compute the ratio, then multiply by $g_0 = 9.81\,\text{m s}^{-2}$.
4. Check: if $h \ll R_E$, verify with $g \approx g_0(1 - 2h/R_E)$.

WORKED EXAMPLE 3 **Field strength at the summit of Mount Cameroon and at satellite altitude**

Take $R_E = 6.37e6\,\text{m}$ and $g_0 = 9.81\,\text{m s}^{-2}$.

(a) Find g at the summit of Mount Cameroon, $h = 4.10e3$ m.
(b) Find g at the altitude of a geostationary satellite, $h = 3.58e7$ m.

SOLUTION

(a) Using the ratio method:

$$\frac{g}{g_0} = \left(\frac{R_E}{R_E + h}\right)^2 = \left(\frac{6.37 \times 10^6}{6.37 \times 10^6 + 4.10 \times 10^3}\right)^2 = \left(\frac{6.370}{6.3741}\right)^2 \approx 0.9987.$$

$$g \approx 9.81 \times 0.9987 \approx 9.80\ \text{m s}^{-2}.$$

Check with the approximation: $g \approx g_0(1 - 2h/R_E) = 9.81\left(1 - \frac{2 \times 4.10 \times 10^3}{6.37 \times 10^6}\right) \approx 9.81(1 - 0.00129) \approx 9.80\ \text{m s}^{-2}$. ✓ The decrease is only about $0.013\ \text{m s}^{-2}$: imperceptible in everyday life.

(b) Here h is not small compared with R_E, so the approximation is forbidden:

$$\frac{g}{g_0} = \left(\frac{6.37 \times 10^6}{6.37 \times 10^6 + 3.58 \times 10^7}\right)^2 = \left(\frac{6.37}{42.17}\right)^2 \approx (0.1511)^2 \approx 0.0228.$$

$$g \approx 9.81 \times 0.0228 \approx 0.22\ \text{m s}^{-2}.$$

At geostationary altitude the Earth's pull is only about 2.3% of its surface value — yet it is exactly enough to keep the satellite in orbit, as we shall see in Section 3.

2.6 Variation of g with Depth (Inside the Earth)

What happens if we descend into a deep mine shaft, at depth d below the surface? The distance from the centre is now $r = R_E - d$, which is less than R_E, so one might naively expect g to increase by the inverse-square law. This is wrong, and the reason is instructive.

When the test mass is inside the Earth, the shell of material lying above it (at radii greater than r) pulls it in all directions, and it can be shown that the contributions of a uniform spherical shell cancel exactly: **a uniform shell exerts no net force on a mass inside it**. Only the sphere of radius r below the mass contributes. Assuming the Earth has **uniform density** ρ, this inner mass is

$$M_{\text{in}} = \rho \cdot \frac{4}{3}\pi r^3.$$

Therefore

$$g(r) = \frac{GM_{\text{in}}}{r^2} = \frac{G}{r^2} \cdot \rho\frac{4}{3}\pi r^3 = \frac{4}{3}\pi G\rho\, r.$$

PROPERTY **Variation of g inside the Earth (uniform density model)**

Inside a uniform spherical Earth, the field strength is proportional to the distance from the

centre:

$$g(r) = \frac{4}{3}\pi G\rho\, r \qquad (r \le R_E),$$

or, in terms of the depth $d = R_E - r$ and the surface value g_0:

$$\boxed{g(d) = g_0\left(1 - \frac{d}{R_E}\right)}$$

At the centre ($d = R_E$, $r = 0$): $g = 0$. (The cancellation of the shell's pull is not examinable, but the linear result and its derivation from $M_{\text{in}} = \frac{4}{3}\pi\rho r^3$ are.)

WARNING **g is maximum at the surface, not at the centre**

A very common mistake is to think that "closer to the centre means stronger gravity". In fact, as you descend, there is less and less mass below you and the pull of the shell above cancels out. g decreases linearly with depth and is exactly zero at the centre: a mass placed there is pulled equally in all directions. The maximum value of g is at the surface.

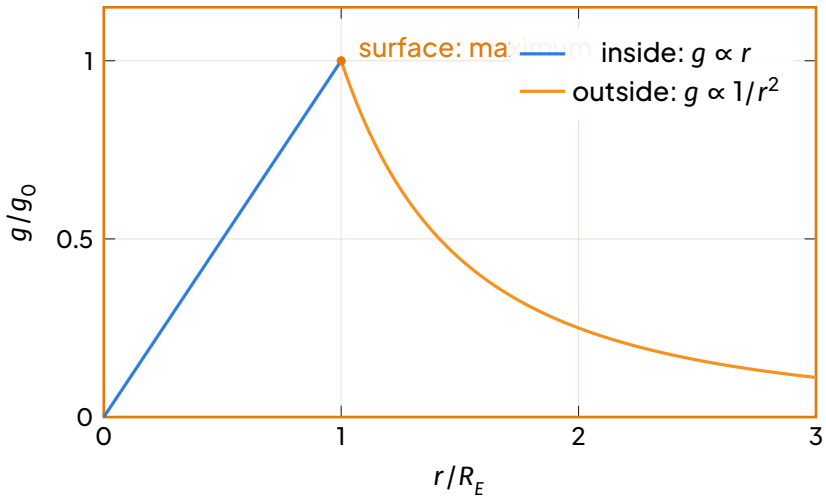
EXEMPLE **g at the centre of the Earth**

At the centre, $d = R_E$, so $g = g_0(1-1) = 0\ \text{m s}^{-2}$. Halfway down ($d = R_E/2$), $g = g_0/2 \approx 4.9\ \text{m s}^{-2}$.

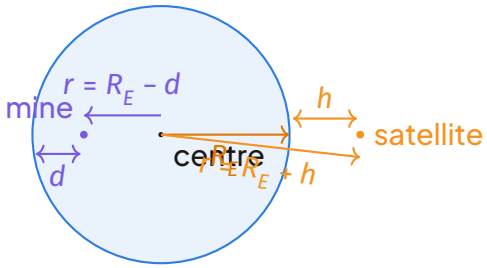
2.7 Graphical Summary: g Against Distance from the Centre

Combining the two results:

$$g(r) = \begin{cases} g_0\,\dfrac{r}{R_E} & \text{for } r \le R_E \quad \text{(linear rise)}, \\ g_0\,\dfrac{R_E^2}{r^2} & \text{for } r \ge R_E \quad \text{(inverse-square fall)}. \end{cases}$$



Variation of g with distance r from the centre of a uniform-density Earth: linear increase inside, inverse-square decrease outside, maximum at the surface $r = R_E$.



altitude h measured upwards from the surface; depth d downwards

Conventions: outside the Earth $r = R_E + h$; inside, $r = R_E - d$. In every formula, r is measured from the centre.

2.8 Effect of the Earth's Rotation and Shape on Apparent g

The value of g actually measured at the surface is not perfectly uniform, for two reasons (qualitative treatment only):

- **Shape.** The Earth is not a perfect sphere: it is slightly flattened at the poles and bulges at the equator, so R_E is slightly larger at the equator. Since $g_0 = GM_E/R_E^2$, the true field is slightly smaller at the equator than at the poles.
- **Rotation.** A body resting on the equator is carried in a circle by the Earth's rotation and needs a small centripetal acceleration $a = \omega^2 R_E$. Part of the gravitational pull is "used up" to provide it, so the apparent weight (what a spring balance reads) is reduced. At the poles there is no such effect.

Both effects act in the same sense: the **apparent g is smallest at the equator and largest at the poles**. The difference is small (a few tenths of a per cent), which is why we usually quote a single average value $g_0 \approx 9.81\,\text{m s}^{-2}$.

DID YOU KNOW? **Weighing less at the equator**

Because of the combined effects of rotation and the equatorial bulge, you would weigh slightly less in Douala (near the equator) than at the poles — even without moving! Precise gravimeters easily detect this difference, and geophysicists use tiny local variations of g to prospect for oil and minerals, since dense underground rocks slightly increase the local field.

WARNING **GCE exam tip**

When a question does not give you G and M_E, do not try to invent them. Express everything in terms of g_0 and R_E using the ratio

$$\frac{g}{g_0} = \frac{R_E^2}{r^2} \quad \text{(outside)}, \qquad \frac{g}{g_0} = \frac{r}{R_E} \quad \text{(inside)}.$$

Also remember: r is always measured from the centre of the Earth, never from the surface. Writing $g = GM/h^2$ with h the altitude is a classic error that costs marks.

EXERCISE 2 **Variation of g**

Take $R_E = 6.37e6\,\text{m}$ and $g_0 = 9.81\,\text{m s}^{-2}$.

1. At what altitude above the surface is g equal to $g_0/4$?
2. A very deep mine reaches a depth $d = 100\,\text{km}$. Using the uniform-density model, find g at the bottom of the mine.
3. Explain, without calculation, why g at the centre of the Earth is zero even though the distance to the centre is zero.

SOLUTION **Exercise 1**

1. Outside the Earth, $\dfrac{g}{g_0} = \dfrac{R_E^2}{(R_E + h)^2} = \dfrac{1}{4}$, so $\dfrac{R_E}{R_E + h} = \dfrac{1}{2}$, giving $R_E + h = 2R_E$, hence

$$h = R_E = 6.37e6\,\text{m}.$$

(Note: g falls to a quarter when the distance from the centre doubles, not the altitude.)

2. Inside the Earth, $g = g_0\left(1 - \frac{d}{R_E}\right) = 9.81\left(1 - \frac{1.0 \times 10^5}{6.37 \times 10^6}\right) = 9.81 \times 0.984 \approx 9.66\ \text{m s}^{-2}$.
3. At the centre, the mass of the Earth surrounds the body symmetrically in all directions. The gravitational pulls from all the surrounding shells cancel each other exactly, so the net force — and therefore $g = F/m$ — is zero. The formula $g = GM/r^2$ does not apply inside the Earth, because the mass is no longer entirely "below" the body.

KEY POINTS **Key points of this section**

- Gravitational field strength: $g = F/m$, unit $\text{N kg}^{-1} = \text{m s}^{-2}$; it is the acceleration of free fall.
- Outside a spherical mass: $g = GM/r^2$ with r measured from the centre; at the surface $g_0 = GM_E/R_E^2$.
- With altitude: $g = g_0 R_E^2/(R_E + h)^2 \approx g_0(1 - 2h/R_E)$ for $h \ll R_E$.
- With depth (uniform density): $g = g_0(1 - d/R_E)$; g falls linearly to zero at the centre.
- g is maximum at the surface; rotation and the equatorial bulge make apparent g smallest at the equator.

3 GRAVITATIONAL POTENTIAL, POTENTIAL ENERGY, ESCAPE AND ORBITAL SPEEDS

In the previous section we described the Earth's gravitational field through the **field strength** $\vec{g}$, a force per unit mass. Force is not the only way to describe a field: it is often far simpler to work with energy. Just as knowing the height of a hill tells you how much energy a cyclist needs to climb it, knowing the **gravitational potential** at a point tells you how much energy a mass needs to get there. In this section we build the energy description of gravitation, and we use it to answer three questions of great practical importance: how fast must a rocket go to escape the Earth, how fast must a satellite go to stay in orbit, and where must a satellite be placed so that it appears fixed above Yaoundé or Douala?

3.1 Work Done by a Gravitational Force: a Conservative Field

EXPERIMENT **Lifting a mass away from the Earth**

Imagine a small mass m resting on the ground. To raise it slowly to a height h you must pull it upwards against the Earth's attraction: you do positive work, and the gravitational force

does negative work. If you then release the mass, gravity pulls it back down and returns exactly the energy you stored. No energy is lost along the way. This is the signature of a **conservative force**: the work it does depends only on the starting and finishing points, never on the path taken.

Consider a mass m moved from point A to point B in the field of a mass M (the Earth). The gravitational force $\vec{F} = -\frac{GMm}{r^2}\,\vec{e}_r$ varies with distance, so the work must be computed by integration along a radial path:

$$W_{A\to B} = \int_A^B \vec{F}\cdot d\vec{r} = \int_{r_A}^{r_B} -\frac{GMm}{r^2}\,dr = GMm\left(\frac{1}{r_B} - \frac{1}{r_A}\right).$$

The result depends only on r_A and r_B. Whatever path is chosen — straight, curved, zig-zag — the work is the same, and the work over any closed path is zero.

PROPERTY **The gravitational field is conservative**

The gravitational force is a **conservative force**: its work between two points is independent of the path followed. Consequently a potential energy E_p can be defined such that

$$W_{A\to B}(\vec{F}_{\text{grav}}) = -\Delta E_p = E_{p,A} - E_{p,B},$$

and the total mechanical energy $E = E_k + E_p$ of a body moving under gravity alone is conserved.

3.2 Gravitational Potential Energy

To define potential energy we must choose a reference level where $E_p = 0$. For objects near the ground we used "$E_p = 0$ at the Earth's surface", but for satellites and rockets the natural choice is a point infinitely far away, where the gravitational force vanishes. Bringing a mass m from infinity to a distance r from M, the work done by gravity is

$$W_{\infty\to r} = \frac{GMm}{r} - 0 = \frac{GMm}{r} = -\big(E_p(r) - E_p(\infty)\big),$$

so, with $E_p(\infty) = 0$:

DEFINITION **Gravitational potential energy**

The **gravitational potential energy** of a mass m at a distance r from the centre of a spherical mass M is

$$\boxed{E_p = -\frac{GMm}{r}} \qquad \text{(zero chosen at infinity).}$$

E_p is measured in joules (J) and is a property of the system (M, m), not of m alone.

Why the negative sign? The minus sign is not a mathematical accident; it carries deep physical meaning. At infinity the system has $E_p = 0$ and the two bodies are free of each other. Any finite separation gives $E_p < 0$: the system has less energy than a free system. To separate the bodies to infinity you must supply energy $\frac{GMm}{r}$. We say the mass m is **bound** to M, trapped in a "potential well". A satellite in orbit, the Moon around the Earth, the Earth around the Sun: all have negative total energy, all are bound.

WARNING **Potential energy is negative — do not "correct" it**

Many students drop the minus sign in $E_p = -GMm/r$ because "energy cannot be negative". It can! Only differences of potential energy are physical, and the zero is a convention. A satellite in orbit is bound: raising it to a higher orbit (larger r) makes E_p less negative, which requires positive work from the rocket engines. Never write $E_p = +GMm/r$.

3.3 Gravitational Potential

Potential energy depends on the test mass m. To characterise the field itself, at each point of space, we divide by m — exactly as we divided force by mass to obtain g.

DEFINITION **Gravitational potential**

The **gravitational potential** at a distance r from the centre of a spherical mass M is the potential energy per unit mass placed at that point:

$$\boxed{V = \frac{E_p}{m} = -\frac{GM}{r}} \qquad \text{unit: J kg}^{-1}.$$

The zero of potential is at infinity. Potentials from several masses add algebraically (scalars — no vector addition needed!).

The **potential difference** between two points gives the work directly: moving a mass m from A to B requires an external agent to supply

$$W_{\text{ext}} = \Delta E_p = m\,\Delta V = m\left(V_B - V_A\right).$$

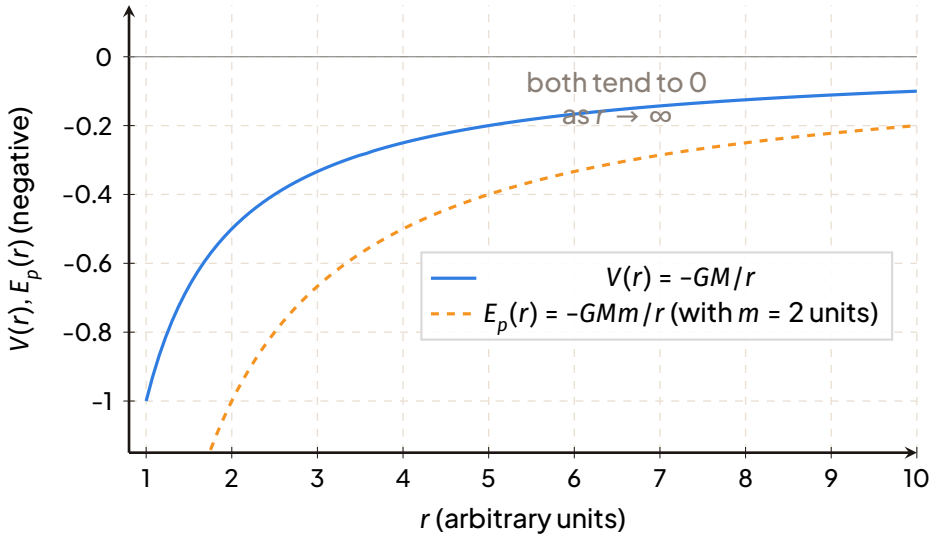
EXEMPLE **Potential at the Earth's surface**

With $M_E = 5.98 \times 10^{24}$ kg, $R_E = 6.37 \times 10^{6}$ m:

$$V_{\text{surface}} = -\frac{GM_E}{R_E} = -\frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24}}{6.37 \times 10^{6}} \approx -6.26 \times 10^{7}\ \text{J kg}^{-1}.$$

Every kilogramme sitting on the ground is "down" by about 63 MJ compared with a kilogramme at infinity. That is the depth of the Earth's potential well.

Equipotential surfaces. Around a spherical mass, V depends only on r, so the surfaces of constant potential are concentric spheres centred on M. The field lines (radial, pointing inward) are everywhere perpendicular to the equipotentials, and no work is needed to move a mass along an equipotential surface. Close to the Earth the equipotentials are closely spaced (strong field); far away they spread apart (weak field).



Gravitational potential $V(r)$ and potential energy $E_p(r)$: both are negative everywhere and rise towards zero at infinity. The system is bound inside this "potential well".

3.4 Variation of Potential Inside the Earth (Extension)

The formulae $V = -GM/r$ and $g = GM/r^2$ apply outside a spherical mass ($r \geq R$). Inside a uniform sphere of radius R and mass M, the field strength grows linearly with r and the potential is parabolic:

PROPERTY **Field and potential inside a uniform sphere**

For $r \leq R$ (inside a uniform sphere of mass M, radius R):

$$g(r) = \frac{GM}{R^3}\,r, \qquad V(r) = -\frac{GM}{2R^3}\left(3R^2 - r^2\right).$$

At the centre ($r = 0$), $g = 0$ and $V_{\text{centre}} = -\dfrac{3GM}{2R} = \dfrac{3}{2}V_{\text{surface}}$. The potential is continuous and smooth at $r = R$.

WARNING **Syllabus focus**

The GCE Advanced Level examination focuses almost exclusively on the external field ($r \geq R$). The internal formulae above are provided for completeness and deeper understanding; they are rarely examined directly.

3.5 Escape Velocity

Intuition. Throw a stone upwards: it falls back. Throw it faster: it rises higher before falling back. Is there a speed for which it never falls back? Yes — the **escape velocity**, the minimum launch speed for which the projectile just reaches infinity with zero speed left.

PROPERTY **Escape velocity — proof examinable**

The **escape velocity** from the surface of a spherical body of mass M and radius R is

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2g_0 R}$$

where $g_0 = GM/R^2$ is the field strength at the surface.

Derivation (energy conservation). Launch a body of mass m from the surface with speed v_e, ignoring air resistance. Mechanical energy is conserved:

$$E_{\text{surface}} = \tfrac{1}{2}mv_e^2 - \frac{GMm}{R}.$$

The minimum escape condition is that the body arrives at infinity with zero kinetic energy (any leftover speed would mean we launched too fast). At infinity $E_k = 0$ and $E_p = 0$, so $E_\infty = 0$. Conservation gives

$$\tfrac{1}{2}mv_e^2 - \frac{GMm}{R} = 0 \quad \Longrightarrow \quad v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2g_0 R}.$$

EXEMPLE **Escape velocity of the Earth**

$$v_e = \sqrt{2g_0 R_E} = \sqrt{2 \times 9.81 \times 6.37 \times 10^6} \approx 1.12 \times 10^4 \,\text{m s}^{-1} \approx 11.2 \,\text{km s}^{-1}.$$

Note that v_e is independent of the mass of the escaping object: a stone and a rocket need the same speed (but very different energies).

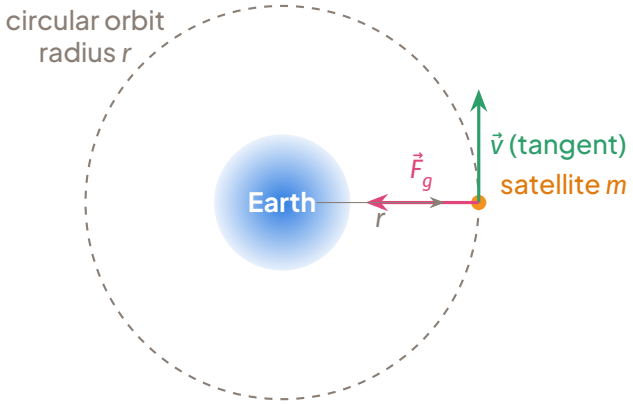
DID YOU KNOW? **Why the Moon has no atmosphere**

Escape velocity depends on M and R. For the Moon, $v_e \approx 2.4 \,\text{km s}^{-1}$, low enough that the fastest molecules of any gas can leak away over geological time. The Earth, with $v_e \approx$

11.2 km s^{-1}, comfortably retains its nitrogen and oxygen. The same physics explains why hydrogen, the lightest and fastest gas, is rare in our atmosphere.

3.6 Satellites in Circular Orbit: Speed, Period, Energy

Setting up the motion. A satellite of mass m in a circular orbit of radius r around the Earth (mass M_E) is in uniform circular motion: the only force is gravity, which plays the role of the centripetal force.



A satellite in circular orbit: the gravitational force $\vec{F}_g$ points to the centre and provides the centripetal force; the velocity $\vec{v}$ is tangent to the orbit.

Newton's second law projected on the radial (centripetal) direction gives

$$\frac{GM_E m}{r^2} = \frac{mv^2}{r} \quad \Longrightarrow \quad \boxed{v = \sqrt{\frac{GM_E}{r}}}.$$

The orbital speed depends only on r: the higher the orbit, the slower the satellite. The period follows from $T = 2\pi r/v$:

$$\boxed{T = 2\pi\sqrt{\frac{r^3}{GM_E}}} \quad \Longleftrightarrow \quad \frac{T^2}{r^3} = \frac{4\pi^2}{GM_E} \text{ (constant)},$$

which is exactly Kepler's third law, now derived rather than merely observed.

PROPERTY **Energy of a circular orbit**

For a satellite in a circular orbit of radius r:

$$E_k = \frac{GM_E m}{2r}, \qquad E_p = -\frac{GM_E m}{r}, \qquad \boxed{E = E_k + E_p = -\frac{GM_E m}{2r} = -E_k = \frac{E_p}{2}}.$$

The total energy is negative: the satellite is bound. To move it to a higher orbit (larger r, energy less negative), the engines must do positive work $\Delta E = \dfrac{GM_E m}{2}\left(\dfrac{1}{r_1} - \dfrac{1}{r_2}\right)$.

Proof. From the orbit condition, $mv^2 = GM_E m/r$, so $E_k = \tfrac{1}{2}mv^2 = \dfrac{GM_E m}{2r}$. Adding $E_p = -\dfrac{GM_E m}{r}$ gives $E = -\dfrac{GM_E m}{2r}$. □

WORKED EXAMPLE 4 **A low-orbit observation satellite**

A satellite of mass $m = 800\ \text{kg}$ circles the Earth at an altitude $h = 600\ \text{km}$. Take $M_E = 5.98 \times 10^{24}\ \text{kg}$, $R_E = 6.37 \times 10^6\ \text{m}$. Find (a) its orbital speed, (b) its period, (c) its total mechanical energy, (d) the energy needed to raise it to an altitude of 1000 km.

SOLUTION

The orbit radius is $r = R_E + h = 6.37 \times 10^6 + 0.600 \times 10^6 = 6.97 \times 10^6\ \text{m}$.
(a) Speed:

$$v = \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24}}{6.97 \times 10^6}} \approx 7.57 \times 10^3\ \text{m s}^{-1} \approx 7.6\ \text{km s}^{-1}.$$

(b) Period:

$$T = 2\pi\sqrt{\frac{r^3}{GM_E}} = 2\pi\sqrt{\frac{(6.97 \times 10^6)^3}{3.99 \times 10^{14}}} \approx 5.80 \times 10^3\ \text{s} \approx 96.6\ \text{min}.$$

(c) Total energy:

$$E = -\frac{GM_E m}{2r} = -\frac{3.99 \times 10^{14} \times 800}{2 \times 6.97 \times 10^6} \approx -2.29 \times 10^{10}\ \text{J}.$$

Negative, as expected for a bound satellite.
(d) Energy to raise the orbit to $r_2 = 7.37 \times 10^6\ \text{m}$:

$$\Delta E = \frac{GM_E m}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right) = \frac{3.99 \times 10^{14} \times 800}{2}\left(\frac{1}{6.97 \times 10^6} - \frac{1}{7.37 \times 10^6}\right) \approx 1.24 \times 10^9\ \text{J}.$$

About 1.2 GJ must be supplied by the engines — positive work, because the new orbit is less tightly bound.

3.7 Geostationary Satellites

The idea. A **geostationary satellite** appears to hang motionless above one point of the equator. Television dishes in Yaoundé, Buea or Bamenda can therefore be bolted in a fixed direction and still receive CRTV or Canal+ relays day and night. For this to happen, three conditions must hold simultaneously:

1. the orbit is circular and lies in the equatorial plane;
2. the period equals the Earth's rotation period, $T = 24\ \text{h}$ (more precisely one sidereal day, $\approx 86\,164\ \text{s}$);
3. the satellite revolves in the same sense as the Earth's rotation (west to east).

METHOD **Finding the geostationary radius**

1. Write Kepler's third law for the orbit: $T^2 = \dfrac{4\pi^2 r^3}{GM_E}$.
2. Impose the geostationary condition $T = 24\ \text{h} = 8.64 \times 10^4\ \text{s}$ (or $86\,164\ \text{s}$ for the sidereal day).
3. Solve for r: $r = \left(\dfrac{GM_E T^2}{4\pi^2}\right)^{1/3}$.
4. Subtract the Earth's radius to get the altitude: $h = r - R_E$.

EXEMPLE **Radius and altitude of the geostationary orbit**

Using $T = 8.64 \times 10^4\ \text{s}$:

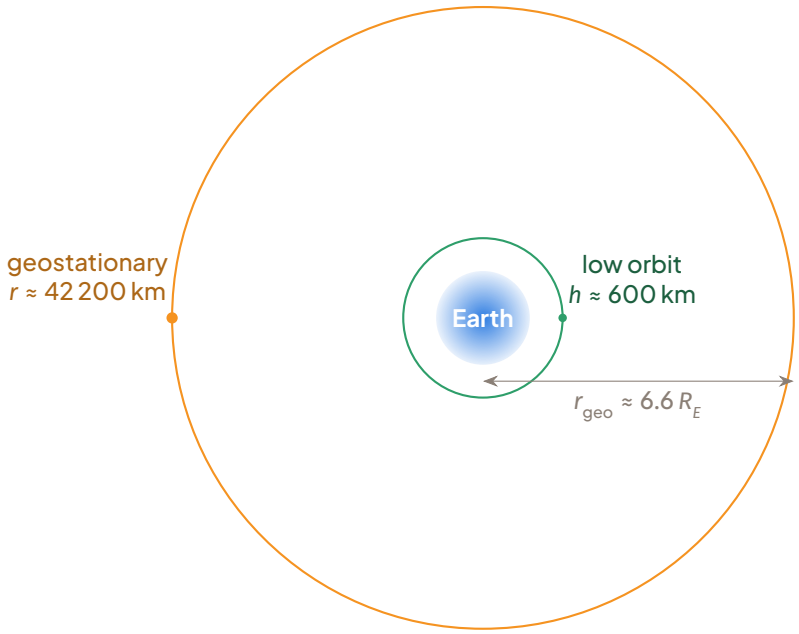
$$r = \left(\frac{3.99 \times 10^{14} \times (8.64 \times 10^4)^2}{4\pi^2}\right)^{1/3} \approx 4.22 \times 10^7\ \text{m} \approx 42\,200\ \text{km},$$

$$h = r - R_E \approx 42\,200 - 6\,370 \approx 35\,800\ \text{km}.$$

(Using the sidereal day $T = 86\,164\ \text{s}$ gives the accepted precise value $h \approx 35\,786\ \text{km}$.) The orbital speed there is

$$v = \sqrt{\frac{GM_E}{r}} \approx 3.07 \times 10^3\ \text{m s}^{-1} \approx 3.1\ \text{km s}^{-1},$$

much slower than a low-orbit satellite, exactly as $v = \sqrt{GM/r}$ predicts.



To scale: a low Earth orbit ($r \approx 1.1\,R_E$) compared with the geostationary orbit ($r \approx 6.6\,R_E$).

DID YOU KNOW? **Why above the equator? — enrichment beyond the strict syllabus**

The centre of any satellite's orbit must coincide with the centre of the Earth, because gravity pulls towards that centre. An orbit tilted relative to the equator is perfectly possible, but its ground track drifts north and south during the day: seen from the ground the satellite traces a figure of eight and is not stationary. Only an orbit in the equatorial plane keeps the satellite permanently above one fixed point. A consequence: geostationary satellites sit on the equatorial ring, so they see the polar regions only at a very grazing angle and cannot serve them properly. Polar observation and high-latitude communications use inclined or polar low orbits instead — as do GPS-type constellations, whose medium-altitude satellites ($T \approx 12$ h) are not geostationary at all.

WARNING **Frequent examination traps**

- Confusing altitude h with orbit radius $r = R_E + h$. All formulas (v, T, E) use r, never h.
- Forgetting to convert the period to seconds before using Kepler's third law.
- Writing $E_k = -E$ as $E_k = +\frac{GMm}{2r}$ but then claiming total energy is positive: for any bound orbit, $E < 0$.
- Believing a geostationary satellite is "weightless because far away": at $r = 42\,200$ km,

$g \approx 0.22\ \text{m s}^{-2}$, small but not zero — it is precisely this force that holds the orbit.

- Forgetting that escape velocity is independent of launch direction (in the absence of atmosphere); only the speed matters.

EXERCISE 3 **Between two orbits**

A 500 kg satellite is transferred from a circular orbit of radius $r_1 = 7.0 \times 10^6$ m to the geostationary radius $r_2 = 4.22 \times 10^7$ m. Using $GM_E = 3.99 \times 10^{14}\ \text{m}^3\ \text{s}^{-2}$:

1. Compute its speed on each orbit.
2. Compute its total mechanical energy on each orbit.
3. Deduce the work the engines must supply. Comment on its sign.

SOLUTION **Exercise 1**

1. $v_1 = \sqrt{GM_E/r_1} = \sqrt{3.99 \times 10^{14}/7.0 \times 10^6} \approx 7.55 \times 10^3\ \text{m s}^{-1}$; $v_2 = \sqrt{3.99 \times 10^{14}/4.22 \times 10^7} \approx 3.07 \times 10^3\ \text{m s}^{-1}$. The higher orbit is slower.
2. $E_1 = -\dfrac{GM_E m}{2r_1} = -\dfrac{3.99 \times 10^{14} \times 500}{1.4 \times 10^7} \approx -1.43 \times 10^{10}$ J; $E_2 = -\dfrac{3.99 \times 10^{14} \times 500}{8.44 \times 10^7} \approx -2.36 \times 10^9$ J.
3. $W = \Delta E = E_2 - E_1 \approx -2.36 \times 10^9 - (-1.43 \times 10^{10}) \approx +1.19 \times 10^{10}$ J. The work is positive: raising a satellite to a less bound (higher) orbit always costs energy, even though the satellite ends up moving slower — the gain in potential energy exceeds the loss in kinetic energy.

KEY POINTS **Key points of the section**

- Gravity is conservative: $E = E_k + E_p$ is conserved, with $E_p = -\dfrac{GMm}{r}$ and zero at infinity. Negative E_p means a bound system.
- Gravitational potential: $V = -\dfrac{GM}{r}$ (in J kg^{-1}), with $E_p = mV$ and $W_{\text{ext}} = m\,\Delta V$.
- Escape velocity: $v_e = \sqrt{\dfrac{2GM}{R}} = \sqrt{2g_0 R} \approx 11.2\ \text{km s}^{-1}$ for the Earth.
- Circular orbit: $v = \sqrt{\dfrac{GM}{r}}$, $T = 2\pi\sqrt{\dfrac{r^3}{GM}}$, $E = -\dfrac{GMm}{2r} = -E_k = \dfrac{E_p}{2}$.
- Geostationary orbit: equatorial, $T = 24$ h (sidereal), same sense as the Earth's rotation; $r \approx 42\,200$ km, $h \approx 35\,786$ km.

4 MOTION IN THE GRAVITATIONAL FIELD: PROJECTILE MOTION

In the previous sections we treated gravity on a grand scale: potentials, escape speeds and satellites orbiting the whole Earth. There, the field was radial and g varied significantly with distance. We now zoom in to a completely different scale: a football kicked on a field in Buea, a stone thrown across a stream, a mango falling from a tree. Over distances of a few tens of metres, the curvature of the Earth is utterly negligible and the gravitational field can be treated as **uniform**. This simplification turns the motion of any thrown object into one of the most elegant and examinable problems in all of mechanics: **projectile motion**.

4.1 The Uniform Field Approximation

DEFINITION **Uniform gravitational field**

A gravitational field is **uniform** over a region if the field vector $\vec{g}$ has the same magnitude and the same direction at every point of that region. Near the Earth's surface, over distances small compared with the Earth's radius R_E, $\vec{g}$ is vertical (pointing towards the centre of the Earth) with magnitude $g \approx 9.81\,\text{m s}^{-2}$.

Why is this legitimate? At altitude h above the surface, $g(h) = \dfrac{GM_E}{(R_E + h)^2}$. For $h = 100\,\text{m}$ and $R_E \approx 6.37e6\,\text{m}$, the relative change in g is about $2h/R_E \approx 3 \times 10^{-5}$ — far below the precision of any school measurement. So for any projectile problem at the GCE, we take $\vec{g}$ constant and vertical.

WARNING **When the uniform model fails**

The uniform-field model is only valid for small-scale motion near the surface. For satellites, intercontinental ballistic motion, or anything rising a significant fraction of R_E, you must return to the inverse-square law of Sections 1–3.

4.2 Setting Up the Equations: Independence of the Motions

Consider a projectile launched from the origin with speed v_0 at angle θ above the horizontal. Choose axes: Ox horizontal in the plane of motion, Oy vertically upward. The only force acting (we neglect air resistance) is the weight $m\vec{g}$, directed along $-Oy$.

Newton's second law gives:

$$m\vec{a} = m\vec{g} \quad \Longrightarrow \quad \vec{a} = \vec{g}$$

so that, component by component:

$$a_x = 0 \qquad \text{and} \qquad a_y = -g.$$

PROPERTY **Independence of horizontal and vertical motions**

In a uniform gravitational field, the motion of a projectile is the superposition of two independent motions:

- a **uniform motion** horizontally: $a_x = 0$, so $v_x = v_0 \cos\theta$ is constant;
- a **uniformly accelerated motion** vertically: $a_y = -g$ (constant).

The horizontal motion does not influence the vertical motion, and vice versa. This is the key that unlocks every projectile problem.

Integrating with the initial conditions $v_x(0) = v_0 \cos\theta$, $v_y(0) = v_0 \sin\theta$, $x(0) = y(0) = 0$:

$$v_x = v_0 \cos\theta, \qquad v_y = v_0 \sin\theta - gt,$$
$$x = (v_0 \cos\theta)\, t, \qquad y = (v_0 \sin\theta)\, t - \tfrac{1}{2} g t^2.$$

EXPERIMENT **Seeing the independence with your own eyes**

Drop one coin vertically from the edge of a table and, at the same instant, flick a second coin horizontally from the same height. Listen: both coins hit the floor at the same time. The horizontal coin travels further, but its fall is identical to that of the dropped coin. Horizontal and vertical motions truly are independent.

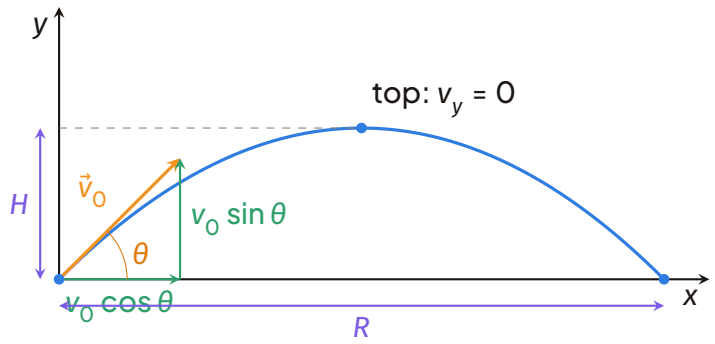
4.3 The Trajectory: a Parabola

Eliminate t between the parametric equations. From $x = (v_0 \cos\theta)t$, we get $t = \dfrac{x}{v_0 \cos\theta}$. Substituting into $y(t)$:

PROPERTY **Trajectory equation**

$$y = x \tan\theta - \frac{g\, x^2}{2 v_0^2 \cos^2\theta}$$

This is of the form $y = ax - bx^2$ with $b > 0$: the trajectory is a **parabola** opening downward.



Parabolic trajectory of a projectile launched at speed v_0 and angle θ. The initial velocity splits into a constant horizontal component $v_0 \cos\theta$ and a vertical component $v_0 \sin\theta$ that gravity steadily reduces.

4.4 Time of Flight, Maximum Height and Range

We now derive the three classic results for a launch **from ground level, landing at the same level**.

Time of flight. Landing means $y = 0$ with $t > 0$:

$$(v_0 \sin\theta)t - \tfrac{1}{2}gt^2 = 0 \implies t\left(v_0 \sin\theta - \tfrac{1}{2}gt\right) = 0 \implies T = \frac{2v_0 \sin\theta}{g}.$$

Maximum height. At the top, $v_y = 0$: $v_0 \sin\theta - gt_{\text{top}} = 0$, so $t_{\text{top}} = \dfrac{v_0 \sin\theta}{g} = \dfrac{T}{2}$ (the motion is symmetric in time). Then

$$H = (v_0 \sin\theta)t_{\text{top}} - \tfrac{1}{2}gt_{\text{top}}^2 = \frac{v_0^2 \sin^2\theta}{2g}.$$

Range. Substituting T into $x = (v_0 \cos\theta)t$:

$$R = (v_0 \cos\theta) \cdot \frac{2v_0 \sin\theta}{g} = \frac{v_0^2 \cdot 2\sin\theta\cos\theta}{g} = \frac{v_0^2 \sin 2\theta}{g}.$$

KEY POINTS **Ground-level launch: the three key results**

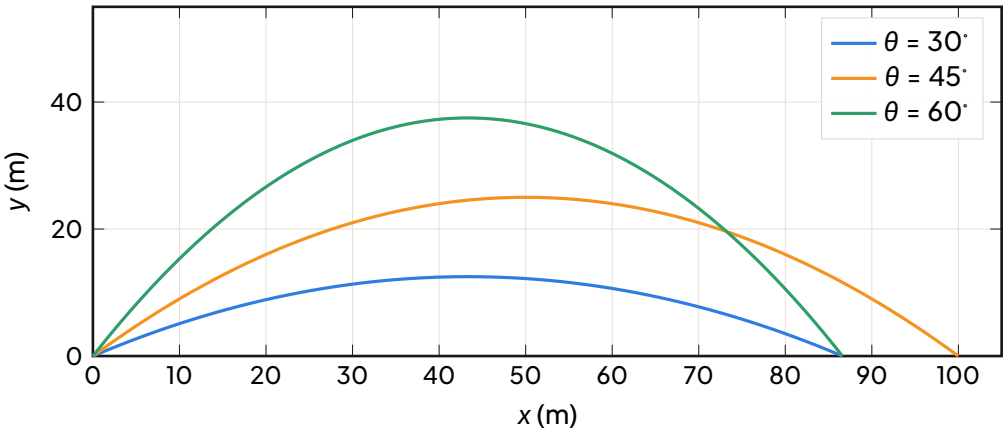
For a projectile launched at speed v_0 and angle θ, landing at the same height:

$$T = \frac{2v_0 \sin\theta}{g}, \qquad H = \frac{v_0^2 \sin^2\theta}{2g}, \qquad R = \frac{v_0^2 \sin 2\theta}{g}.$$

Maximum range and complementary angles. Since $\sin 2\theta \le 1$ with maximum at $2\theta = 90^\circ$:

$$\theta = 45^\circ \implies R_{\max} = \frac{v_0^2}{g}.$$

Because $\sin 2\theta = \sin(180^\circ - 2\theta)$, two complementary angles θ and $90^\circ - \theta$ (e.g. 30° and 60°) give the **same range** — but different times of flight and different maximum heights: the high lob (60°) stays in the air twice as long as the flat throw (30°) for the same landing point.



Trajectories for the same launch speed v_0 (here $v_0^2/g = 100\,\text{m}$) at $\theta = 30^\circ, 45^\circ, 60^\circ$. The 45° launch gives the greatest range; the 30° and 60° launches share the same range but very different heights and flight times.

WARNING **The velocity at the top is NOT zero**

A very common error: "at the highest point the velocity is zero." False. Only the vertical component vanishes: $v_y = 0$. The horizontal component $v_x = v_0 \cos\theta$ is unchanged throughout the flight, so at the top the speed is $v_0 \cos\theta$, directed horizontally. If the velocity were zero, the projectile would simply fall straight down from there!

WARNING **The range formula has a restricted domain of validity**

$R = \dfrac{v_0^2 \sin 2\theta}{g}$ applies only when launch and landing are at the same height. If a ball is kicked from the top of a hill, or lands on a roof, you must go back to the parametric equations and solve $y(t) = h_{\text{landing}}$ for the time, then compute $x(T)$. Blindly applying the formula is a classic source of lost marks.

4.5 Horizontal Launch from a Height

A stone is thrown horizontally with speed v_0 from the top of a cliff of height h (or a ball rolls off a table). This is the special case $\theta = 0^\circ$, with launch point at height h.

Take the origin at the **base of the cliff**, Oy upward. Initial conditions: $x(0) = 0, y(0) = h, v_x(0) = v_0$,

$v_y(0) = 0$. Hence:

$$x = v_0 t, \qquad y = h - \tfrac{1}{2} g t^2.$$

Time of fall: set $y = 0$:

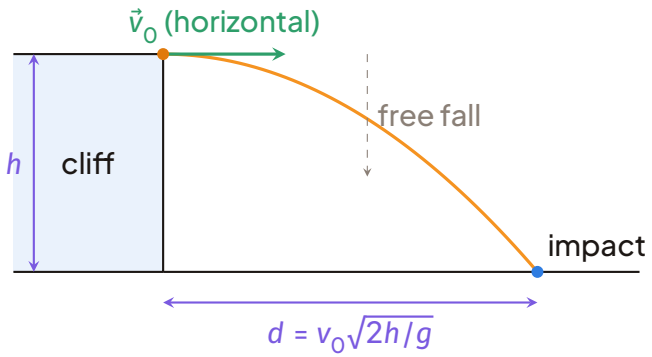
$$t_f = \sqrt{\frac{2h}{g}} \quad \text{— exactly the free-fall time from height } h.$$

The horizontal speed has no effect on the time of fall — the independence principle again.

Horizontal distance (range from the cliff base):

$$d = v_0 \, t_f = v_0 \sqrt{\frac{2h}{g}}.$$

Impact velocity: at impact, $v_x = v_0$ and $v_y = -g t_f = -\sqrt{2gh}$, so the speed is $v = \sqrt{v_0^2 + 2gh}$ — which energy conservation ($\tfrac{1}{2}mv^2 = \tfrac{1}{2}mv_0^2 + mgh$) confirms instantly.



Horizontal launch from a cliff of height h. The vertical motion is pure free fall from rest; the horizontal motion is uniform at speed v_0. The time of fall depends only on h.

WORKED EXAMPLE 5 **Ball rolling off a laboratory table**

A ball rolls horizontally off a table of height $h = 1.25$ m with speed $v_0 = 2.0\ \text{m s}^{-1}$. Take $g = 10\ \text{m s}^{-2}$. Find (a) the time of fall, (b) the horizontal distance from the foot of the table to the impact point, (c) the velocity at impact.

SOLUTION

(a) Time of fall. Vertical motion: $y = h - \tfrac{1}{2} g t^2$. Setting $y = 0$:

$$t_f = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 1.25}{10}} = \sqrt{0.25} = 0.50\ \text{s}.$$

(b) Horizontal distance. Uniform horizontal motion:

$$d = v_0 t_f = 2.0 \times 0.50 = 1.0\ \text{m}.$$

(c) Impact velocity. Components at impact:

$$v_x = v_0 = 2.0\ \text{m s}^{-1}, \qquad v_y = -gt_f = -10 \times 0.50 = -5.0\ \text{m s}^{-1}.$$

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{4.0 + 25} = 5.4\ \text{m s}^{-1}, \qquad \tan\alpha = \frac{|v_y|}{v_x} = 2.5 \Rightarrow \alpha \approx 68^\circ \text{ below the horizontal.}$$

Check with energy: $v = \sqrt{v_0^2 + 2gh} = \sqrt{4 + 25} = 5.4\ \text{m s}^{-1}$. ✓

4.6 Projectiles as Free Fall: the Monkey and the Hunter

Here is a famous piece of reasoning that reveals the deep meaning of projectile motion. A hunter aims directly at a monkey hanging from a branch. At the instant the dart leaves the blowpipe, the monkey lets go and falls. Does the dart hit the monkey?

Yes — always (provided the dart reaches the monkey's vertical line before the ground intervenes). The argument: imagine gravity were switched off. The dart would fly straight along the line of aim and hit the monkey at the aimed point P. Now switch gravity on. In time t, both the dart and the monkey fall the same extra distance $\frac{1}{2}gt^2$ below where they would have been. So at time t the dart is exactly $\frac{1}{2}gt^2$ below P, and the monkey is exactly $\frac{1}{2}gt^2$ below its starting point — which is P. They meet.

KEY POINTS **The deep idea**

A projectile is in **free fall** at every instant: relative to the straight-line path it would follow without gravity, it "falls" by $\frac{1}{2}gt^2$, whatever its launch speed or angle. Gravity accelerates everything identically — dart, monkey, football, satellite.

4.7 From Projectiles to Satellites: Falling Around the Earth

Newton himself proposed a beautiful thought experiment. Fire a projectile horizontally from the top of a very high mountain, above the atmosphere. At modest speed, it follows a parabola (locally) and lands at A. Fire it faster: it travels further before landing, at B, then C. But the Earth is round — its surface curves away from the projectile. There exists a special speed for which the projectile's fall toward the Earth exactly matches the curving away of the Earth's surface: the projectile **never lands**. It is perpetually falling toward the Earth while never getting closer to the ground: it is in **orbit**.

That special speed is precisely the orbital speed of Section 3: $v = \sqrt{\frac{GM_E}{R_E}} \approx 7.9\ \text{km s}^{-1}$ for a low orbit. A satellite is nothing more than a projectile launched fast enough to fall around the Earth

forever. This closes the conceptual loop of the whole chapter: the same inverse-square law that holds the Moon also bends the path of a thrown stone.

DID YOU KNOW? Why astronauts float

Astronauts in the International Space Station are not beyond Earth's gravity — g up there is still about 90% of its surface value. They float because the station and everything in it are in continuous free fall together, exactly like the monkey and the dart. "Weightlessness" is really "fallingness."

4.8 Solving GCE-Style Projectile Problems

METHOD A four-step strategy for any projectile problem

1. Choose axes and state the sign convention. Usually Ox horizontal in the direction of motion, Oy vertically upward, origin at the launch point (or at ground level for cliff problems). Then $a_x = 0$, $a_y = -g$.
2. Resolve the initial velocity. $v_{0x} = v_0 \cos\theta$, $v_{0y} = v_0 \sin\theta$. For a horizontal launch, $v_{0y} = 0$.
3. Write the four equations.

$$v_x = v_{0x}, \quad v_y = v_{0y} - gt, \quad x = v_{0x}t, \quad y = v_{0y}t - \tfrac{1}{2}gt^2.$$

4. Translate the question into a condition and solve. "Highest point" $\Rightarrow v_y = 0$. "Lands" $\Rightarrow y =$ landing height. "Hits a wall" $\Rightarrow x =$ wall distance. Then check units and sign of every answer.

WORKED EXAMPLE 6 GCE-style: the football free kick

A footballer kicks a ball from ground level with speed $v_0 = 20\ \text{m s}^{-1}$ at $\theta = 30^\circ$ above the horizontal. Take $g = 10\ \text{m s}^{-2}$. Determine (a) the time of flight, (b) the maximum height, (c) the range, (d) the speed at the highest point.

SOLUTION

Components: $v_{0x} = 20\cos 30^\circ = 10\sqrt{3} \approx 17.3\ \text{m s}^{-1}$, $v_{0y} = 20\sin 30^\circ = 10\ \text{m s}^{-1}$.

(a) $T = \dfrac{2v_0 \sin\theta}{g} = \dfrac{2 \times 10}{10} = 2.0\ \text{s}.$

(b) $H = \dfrac{v_0^2 \sin^2\theta}{2g} = \dfrac{400 \times 0.25}{20} = 5.0\ \text{m}.$

(c) $R = \dfrac{v_0^2 \sin 2\theta}{g} = \dfrac{400 \times \sin 60^\circ}{10} = 40 \times 0.866 = 35\ \text{m}$ (2 s.f.).
(d) At the top, $v_y = 0$ but v_x persists: $v = v_{0x} = 17.3\ \text{m s}^{-1}$ (horizontal). Note the answer is not zero.

EXERCISE 4 **Practice**

A stone is thrown horizontally with speed $15\ \text{m s}^{-1}$ from the top of a cliff $45\ \text{m}$ high above the sea. Take $g = 10\ \text{m s}^{-2}$ and neglect air resistance.

1. Calculate the time the stone takes to reach the sea.
2. Calculate the horizontal distance from the foot of the cliff to the point of impact.
3. Calculate the speed of the stone just before impact.
4. A second stone is simply dropped from the top of the cliff at the same instant. Which stone hits the sea first? Justify your answer.

SOLUTION **Exercise 1**

1. Vertical motion is free fall from rest: $t_f = \sqrt{2h/g} = \sqrt{2 \times 45/10} = \sqrt{9} = 3.0\ \text{s}$.
2. $d = v_0 t_f = 15 \times 3.0 = 45\ \text{m}$.
3. $v_x = 15\ \text{m s}^{-1}$, $v_y = -gt_f = -30\ \text{m s}^{-1}$, so $v = \sqrt{15^2 + 30^2} = \sqrt{1125} \approx 33.5\ \text{m s}^{-1}$.
4. Both hit the sea at the same time. The vertical motion of the thrown stone is identical to that of the dropped stone (both start with zero vertical velocity and fall $45\ \text{m}$ under gravity). The horizontal speed affects only where it lands, not when.

KEY POINTS **Chapter summary of Section 4**

- Near the Earth's surface, $\vec{g}$ is uniform: constant magnitude $9.81\ \text{m s}^{-2}$, constant (vertical) direction.
- Projectile motion = uniform horizontal motion + uniformly accelerated vertical motion; the two are independent.
- Trajectory: $y = x \tan\theta - \dfrac{gx^2}{2v_0^2 \cos^2\theta}$ — a parabola.
- Same-level launch: $T = \dfrac{2v_0 \sin\theta}{g}$, $H = \dfrac{v_0^2 \sin^2\theta}{2g}$, $R = \dfrac{v_0^2 \sin 2\theta}{g}$; maximum range at $\theta =$

45°; complementary angles give equal ranges.

- Horizontal launch from height h: $t_f = \sqrt{2h/g}$ (independent of v_0), $d = v_0\sqrt{2h/g}$.
- At the top of the path, only v_y is zero; the speed is $v_0 \cos\theta$.
- A satellite is a projectile falling around the Earth — the conceptual bridge between this section and orbital motion.

5 INTEGRATION ACTIVITY

6 INTEGRATION ACTIVITY: DESIGNING A CAMEROONIAN COMMUNICATIONS SATELLITE

6.1 Aims of the Activity

This integration activity closes the chapter Gravitational Fields by asking you to combine, in one authentic engineering-style project, every major result you have studied:

- **Newton's law of gravitation** $F = \frac{GMm}{r^2}$ and the gravitational field strength $g(r) = \frac{GM}{r^2}$;
- **Kepler's third law** $T^2 = \frac{4\pi^2}{GM} r^3$ applied to circular orbits;
- **Orbital speed** $v = \sqrt{\frac{GM}{r}}$ for a circular orbit;
- **Gravitational potential energy** $E_p = -\frac{GMm}{r}$ and energy balances between the Earth's surface and an orbit;
- **Escape velocity** $v_{esc} = \sqrt{\frac{2GM}{R_E}}$ and its physical meaning;
- scientific communication: presenting and defending a numerical design in front of an audience.

Beyond the calculations, the activity trains you to work like real engineers: choosing a model, stating its assumptions, computing with coherent units and sensible significant figures, comparing with accepted values, and discussing the limits of the model.

6.2 Situation

Designing a Cameroonian Communications Satellite

The (fictional) National Agency for Space Applications of Cameroon wishes to provide permanent television broadcasting, telephone relay and internet coverage to the whole national territory, from the forests of the South Region to the savannas of the Far North. The agency's board decides that the most economical solution is a single **geostationary** satellite: a satellite placed in a circular orbit in the plane of the equator, turning in the same sense as the Earth, with a period exactly equal to the Earth's rotation period. Seen from the ground in Yaoundé or Douala, such a satellite appears motionless in the sky, so that fixed parabolic antennas can point at it permanently.

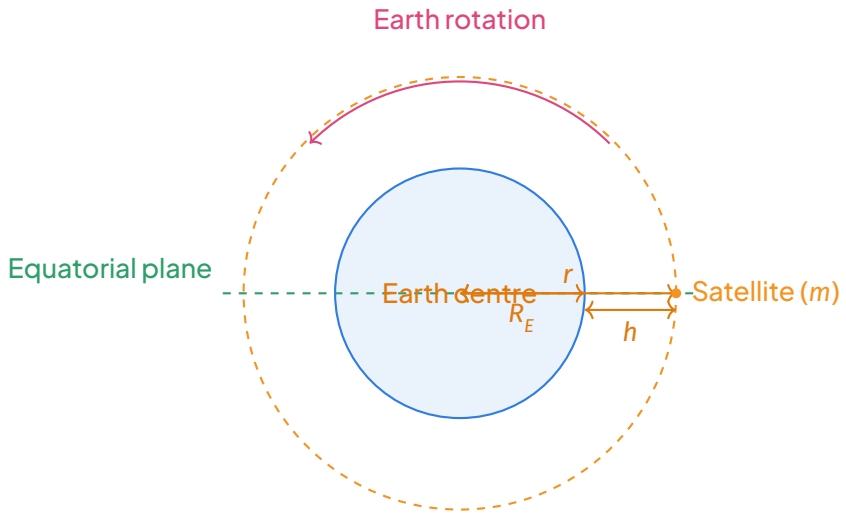
Your class is divided into engineering teams. Each team receives the following mission order from the agency director:

"Determine the radius and altitude of the orbit our satellite must occupy, the speed it must be given there, the strength of the Earth's gravitational field at that altitude, the minimum energy per kilogram needed to raise it from the launch pad to its orbit, and the speed a rocket would need to escape the Earth completely. Present a one-page justified report."

Data provided to all teams

Quantity	Symbol	Value
Gravitational constant	G	$6.67e-11\ \mathrm{N\,m^2\,kg^{-2}}$
Mass of the Earth	M_E	$5.97e24\ \mathrm{kg}$
Mean radius of the Earth	R_E	$6.37e6\ \mathrm{m}$
Sidereal rotation period of the Earth	T	$8.616e4\ \mathrm{s}$ (23 h 56 min)
Mass of the satellite	m	350 kg

Modelling assumptions: the Earth is spherical and non-rotating for the field calculations; the orbit is circular and lies in the equatorial plane; only the Earth's gravity acts on the satellite (the Moon, the Sun and atmospheric drag are neglected); the launch pad is at sea level on the equator.



Geometry of a geostationary orbit: circular, equatorial, radius $r = R_E + h$.

6.3 Tasks

EXERCISE 5 **Mission Order**

1. Orbit radius. Starting from Newton's second law applied to the satellite in uniform circular motion (centripetal force supplied by gravity), derive Kepler's third law $T^2 =$

$\frac{4\pi^2}{GM_E}r^3$. Hence compute the orbit radius r and the altitude $h = r - R_E$ of the geostationary orbit. Express h in kilometres.

2. Orbital speed and field strength. Deduce the orbital speed v of the satellite, in $\mathrm{m\,s^{-1}}$ then in $\mathrm{km\,h^{-1}}$. Compute the gravitational field strength $g(r)$ at the orbit and compare it with $g_0 = 9.81\,\mathrm{m\,s^{-2}}$ at the surface. Is the satellite "weightless"? Comment.

3. Escape velocity. Using an energy argument, derive the expression $v_{\mathrm{esc}} = \sqrt{\frac{2GM_E}{R_E}}$ and compute its value from the Earth's surface. Explain, in a few sentences, why the launcher must give the satellite a speed between the orbital speed and the escape velocity, and why a speed equal to or greater than v_{esc} would be a failure for this mission.

4. Energy cost. Write the gravitational potential energy of the satellite on the launch pad ($r = R_E$) and on its orbit ($r = R_E + h$). Compute the minimum energy per kilogram, $\frac{\Delta E_p}{m}$, required to raise the satellite to its orbit (neglecting the kinetic energy of the orbital motion in a first approach). Then add the kinetic energy per kilogram $\frac{v^2}{2}$ needed on orbit and give the total energy per kilogram. Why is the real energy cost of a launch much higher?

5. Comparison and report. Compare your results with the accepted values (altitude $\approx$ 35786 km, orbital speed $\approx 3.1\,\mathrm{km\,s^{-1}}$). Identify possible sources of discrepancy. Write a one-page report (text, key formulas, one diagram) addressed to the agency director, justifying your design.

6.4 Model Solution

WORKED EXAMPLE 7 **Complete Solution**

1. Orbit radius from Kepler's third law.
For a satellite of mass m in a circular orbit of radius r, the gravitational force provides the necessary centripetal force:

$$\frac{GM_E m}{r^2} = \frac{mv^2}{r}.$$

The orbital speed is $v = \frac{2\pi r}{T}$, where T is the orbital period. Substituting v:

$$\frac{GM_E}{r^2} = \frac{4\pi^2 r}{T^2} \quad \Longrightarrow \quad T^2 = \frac{4\pi^2}{GM_E}r^3,$$

which is Kepler's third law for a circular orbit. For a geostationary satellite, T must equal

the Earth's sidereal rotation period. Solving for r:

$$r = \left(\frac{GM_E T^2}{4\pi^2}\right)^{1/3}.$$

Substitute the data ($G = 6.67e-11\,\mathrm{N\,m^2\,kg^{-2}}$, $M_E = 5.97e24\,\mathrm{kg}$, $T = 8.616e4\,\mathrm{s}$):

$$GM_E = 3.982e14\,\mathrm{m^3\,s^{-2}}, \qquad T^2 = 7.423e9\,\mathrm{s^2}.$$

$$r = \left(\frac{3.982e14\,\mathrm{m^3\,s^{-2}} \times 7.423e9\,\mathrm{s^2}}{4\pi^2}\right)^{1/3} = \left(\frac{2.956e24\,\mathrm{m^3}}{39.48}\right)^{1/3} = \left(7.487e22\,\mathrm{m^3}\right)^{1/3} \approx 4.216e7\,\mathrm{m}.$$

Hence the altitude:

$$h = r - R_E = 4.216e7\,\mathrm{m} - 6.37e6\,\mathrm{m} = 3.579e7\,\mathrm{m} \approx 35790\,\mathrm{km}.$$

KEY POINTS **Result 1**

Orbit radius $r \approx 4.22e7$ m; Altitude $h \approx 35790$ km (3 significant figures).

2. Orbital speed and field strength.
Orbital speed:

$$v = \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{3.982e14\,\mathrm{m^3\,s^{-2}}}{4.216e7\,\mathrm{m}}} = \sqrt{9.445e6\,\mathrm{m^2\,s^{-2}}} \approx 3073\,\mathrm{m\,s^{-1}} \approx 1.11e4\,\mathrm{km\,h^{-1}}.$$

Gravitational field strength at the orbit:

$$g(r) = \frac{GM_E}{r^2} = \frac{3.982e14\,\mathrm{m^3\,s^{-2}}}{(4.216e7\,\mathrm{m})^2} = \frac{3.982e14\,\mathrm{m^3\,s^{-2}}}{1.777e15\,\mathrm{m^2}} \approx 0.224\,\mathrm{m\,s^{-2}}.$$

Comparison: $\frac{g(r)}{g_0} = \frac{0.224}{9.81} \approx 0.0228 \approx 2.3\%$.

WARNING **Weightlessness?**

The satellite is not weightless. The gravitational field at that altitude is 2.3% of the surface value; gravity is exactly what provides the centripetal acceleration keeping the satellite in orbit. The sensation of weightlessness inside the satellite arises because the satellite and everything in it are in **free fall** (they accelerate towards Earth at exactly $g(r)$).

3. Escape velocity.
To escape to infinity with zero residual kinetic energy, the total mechanical energy at the surface must be at least zero:

$$E_{\text{total}} = \frac{1}{2}mv_{\text{esc}}^2 - \frac{GM_E m}{R_E} \geq 0.$$

The minimum escape velocity corresponds to $E_{\text{total}} = 0$:

$$\frac{1}{2}mv_{\text{esc}}^2 = \frac{GM_E m}{R_E} \quad \Longrightarrow \quad v_{\text{esc}} = \sqrt{\frac{2GM_E}{R_E}}.$$

Numerically:

$$v_{\text{esc}} = \sqrt{\frac{2 \times 3.982e14\,\text{m}^3\,\text{s}^{-2}}{6.37e6\,\text{m}}} = \sqrt{1.250e8\,\text{m}^2\,\text{s}^{-2}} \approx 11180\,\text{m}\,\text{s}^{-1} \approx 11.2\,\text{km}\,\text{s}^{-1}.$$

KEY POINTS **Result 3**

$v_{\text{esc}} \approx 11.2\,\text{km}\,\text{s}^{-1}$, while the orbital speed is $v \approx 3.07\,\text{km}\,\text{s}^{-1}$.

The launcher must impart a speed greater than the low-orbit circular speed to reach the geostationary altitude, but strictly less than v_{esc}. If $v \geq v_{\text{esc}}$, the satellite would have enough energy to leave the Earth's gravitational influence entirely and would never enter the desired closed orbit — the mission would fail. The useful injection window lies between the transfer-orbit perigee speed and the escape speed.

4. Energy cost per kilogram.

Gravitational potential energy per unit mass:

$$\frac{E_p}{m} = -\frac{GM_E}{r}.$$

At the surface ($r = R_E$):

$$\frac{E_{p,\text{surf}}}{m} = -\frac{3.982e14\,\text{m}^3\,\text{s}^{-2}}{6.37e6\,\text{m}} = -6.251e7\,\text{J}\,\text{kg}^{-1}.$$

On the geostationary orbit ($r = 4.216e7\,\text{m}$):

$$\frac{E_{p,\text{orb}}}{m} = -\frac{3.982e14\,\text{m}^3\,\text{s}^{-2}}{4.216e7\,\text{m}} = -9.445e6\,\text{J}\,\text{kg}^{-1}.$$

Minimum potential energy gain per kilogram (work against gravity):

$$\frac{\Delta E_p}{m} = \frac{E_{p,\text{orb}} - E_{p,\text{surf}}}{m} = GM_E\left(\frac{1}{R_E} - \frac{1}{r}\right) = 6.251e7\,\text{J}\,\text{kg}^{-1} - 9.445e6\,\text{J}\,\text{kg}^{-1} \approx 5.307e7\,\text{J}\,\text{kg}^{-1}.$$

Kinetic energy per kilogram required on the circular orbit:

$$\frac{E_{k,\text{orb}}}{m} = \frac{v^2}{2} = \frac{(3073\,\text{m}\,\text{s}^{-1})^2}{2} \approx 4.72e6\,\text{J}\,\text{kg}^{-1}.$$

Total minimum mechanical energy per kilogram (ideal, no losses):

$$\frac{E_{\text{total}}}{m} = \frac{\Delta E_p}{m} + \frac{E_{k,\text{orb}}}{m} \approx 5.307e7\,\text{J}\,\text{kg}^{-1} + 4.72e6\,\text{J}\,\text{kg}^{-1} = 5.78e7\,\text{J}\,\text{kg}^{-1}.$$

For the 350 kg satellite, this represents ≈ 2.02e10 J.

WARNING **Real launch cost**

The real energy cost is far higher because:

- The rocket must lift its own structure, engines, tanks and the vast majority of its mass as propellant (the **tyranny of the rocket equation**).
- Atmospheric drag during the first ~ 100 km dissipates energy.
- No engine is perfectly efficient (thermal losses, incomplete combustion).
- Gravity losses: thrust spent fighting gravity while not yet at orbital speed.
- Steering and plane-change manoeuvres.

Typical launchers require ~ 30 MJ kg^{-1} to 50 MJ kg^{-1} of payload to GEO, i.e., 50–100 times the theoretical minimum.

5. Comparison, discussion and report outline.
Our computed values:

- Altitude h = 35790 km vs. accepted 35786 km (difference < 0.02%).
- Orbital speed v = 3.07 km s^{-1} vs. accepted 3.07 km s^{-1} (excellent agreement).

Sources of the tiny residual discrepancies:

- Rounded values of G, M_E, R_E (3 significant figures only).
- Use of the sidereal day (86164 s) rather than the solar day (86400 s) — correct for geostationary orbit.
- Spherical Earth model (Earth's equatorial radius is 6378 km, polar radius 6357 km; J_2 oblateness perturbs the orbit slightly).
- Neglect of lunar/solar perturbations and solar radiation pressure.

METHOD **One-page report structure for the Agency Director**

1. Title: Geostationary Orbit Parameters for Cameroon Communications Satellite.
2. Summary: One sentence giving $h, v, g(r), \Delta E/m, v_{esc}$.
3. Assumptions: Spherical Earth, equatorial circular orbit, only Earth gravity.
4. Key formulas: $T^2 = 4\pi^2 r^3/(GM_E)$, $v = \sqrt{GM_E/r}$, $g = GM_E/r^2$, $v_{esc} = \sqrt{2GM_E/R_E}$, $\Delta E/m = GM_E(1/R_E - 1/r) + v^2/2$.
5. Results table: Symbol, Formula, Value, Unit.
6. Diagram: Earth, orbit, R_E, h, r, satellite, equatorial plane (as in the figure above).
7. Conclusion: Orbit validated; energy budget estimated; recommendation for launcher selection (e.g., Ariane 5/6, Falcon 9, Long March 3B capabilities).

WARNING **Traps to Avoid in This Activity**

- Do not confuse **radius** r (from the Earth's centre) with **altitude** h (from the surface): every formula of this chapter uses r.
- Convert the period to seconds before substituting; mixing hours and metres is the most common source of absurd results.
- In Kepler's third law, T is the sidereal day ($8.616e4$ s), not 24 h: a geostationary satellite must turn with the Earth relative to the stars.
- Potential energies are negative; the energy to supply is the positive difference $\Delta E_p = E_{p,orb} - E_{p,surf}$.
- Escape velocity is derived from energy conservation, not from force balance.

DID YOU KNOW? **Cameroon and Space Applications**

Cameroon does not yet have its own launch capability, but it is an active user of space services. The country hosts ground stations for satellite telemetry (e.g., in Kribi) and benefits from geostationary satellites like Intelsat, Eutelsat, RASCOM-QAF1 (the first pan-African satellite, launched 2007) and Belintersat-1 for broadcasting, VSAT internet, and telephony.

The African Space Agency (headquartered in Cairo, Egypt, since 2023) coordinates continental space policy. A future Cameroonian geostationary satellite would likely be procured from a manufacturer (Airbus, Thales Alenia Space, CAST) and launched on a commercial rocket (Arianespace, SpaceX, CNSA), exactly as modelled in this activity.

KEY POINTS **Link to Projectile Motion (Lower Sixth Mechanics)**

The trajectory of a projectile near Earth's surface is a parabola (constant g approximation). The trajectory of a satellite is a conic section (ellipse, parabola, hyperbola) governed by the inverse-square law. A geostationary orbit is a circular ellipse ($e = 0$). An escape trajectory is a parabola ($e = 1$, total energy = 0). A hyperbolic trajectory ($e > 1$, total energy > 0) corresponds to interplanetary transfer. The unification of projectile and orbital motion under Newton's law of gravitation is one of the great syntheses of classical mechanics.

7 METHODS AND WORKED EXAMPLES

8 GRAVITATIONAL FIELDS

8.1 Newton's Law of Universal Gravitation and Kepler's Laws

DEFINITION **Newton's Law of Universal Gravitation**

Every particle of matter in the universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres. The force acts along the line joining the centres and is always attractive.

PROPERTY **Mathematical Form of Newton's Law**

For two point masses m_1 and m_2 separated by a distance r (centre-to-centre), the magnitude of the gravitational force is

$$F = \frac{G\,m_1 m_2}{r^2},$$

where $G = 6.67 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$ is the universal gravitational constant. In vector form, the force on m_2 due to m_1 is

$$\vec{F}_{12} = -\frac{G\,m_1 m_2}{r^2}\,\hat{r}_{12},$$

where $\hat{r}_{12}$ is the unit vector from m_1 to m_2. The force on m_1 due to m_2 is $\vec{F}_{21} = -\vec{F}_{12}$ (Newton's third law).

KEY POINTS **Key Points for Newton's Law**

- G is a universal constant; its value is the same everywhere in the universe.
- The law applies exactly to point masses and to spherically symmetric bodies (where all mass can be considered concentrated at the centre).
- For non-spherical or non-symmetric bodies, the law must be applied by integration over mass elements.
- Gravitational force is a **central force** (always directed along the line joining centres) and a **conservative force**.

- The inverse-square nature means: if distance doubles, force becomes one-quarter; if distance triples, force becomes one-ninth.

METHOD **Calculating a gravitational force between two bodies**

1. Identify the two masses m_1, m_2 and the centre-to-centre distance r (never the surface-to-surface distance!). Convert everything to SI units (kg, m).
2. Write Newton's law:

$$F = \frac{G m_1 m_2}{r^2}, \qquad G = 6.67 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}.$$

3. Compute the magnitude. The force is always **attractive** and acts along the line joining the centres.
4. For extended spherical bodies (planets, stars), treat all the mass as concentrated at the centre — this is exact for spherically symmetric bodies (Shell Theorem).
5. Check the order of magnitude: gravitational forces between laboratory-sized objects are tiny (10^{-10} to 10^{-7} N); they become large only when at least one mass is astronomical.

Reflex: if the distance is doubled, the force is divided by 4 (inverse-square law). Use ratios to avoid heavy computation.

WORKED EXAMPLE 8 **Force between the Earth and the Moon**

The Earth has mass $M_E = 5.98 \times 10^{24}$ kg and the Moon has mass $M_M = 7.35 \times 10^{22}$ kg. Their centres are $r = 3.84 \times 10^{8}$ m apart.

1. Calculate the gravitational force exerted by the Earth on the Moon.
2. State, with justification, the force exerted by the Moon on the Earth.
3. Show that this force provides exactly the centripetal force needed if the Moon orbits the Earth with period $T = 27.3$ days.

SOLUTION

1. Force exerted by the Earth on the Moon.
Both bodies are nearly spherical, so we apply Newton's law with the centre-to-centre distance:

$$F = \frac{GM_E M_M}{r^2} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(7.35 \times 10^{22})}{(3.84 \times 10^{8})^2}.$$

Numerator: $(6.67 \times 10^{-11})(5.98 \times 10^{24}) = 3.989 \times 10^{14}$; then $(3.989 \times 10^{14})(7.35 \times 10^{22}) = 2.932 \times 10^{37}\ \text{N m}^2$. Denominator: $(3.84 \times 10^{8})^2 = 1.4746 \times 10^{17}\ \text{m}^2$.

$$F = \frac{2.932 \times 10^{37}}{1.4746 \times 10^{17}} = 1.99 \times 10^{20}\ \text{N}.$$

The force is attractive, directed from the Moon towards the centre of the Earth.

2. Force of the Moon on the Earth.

By Newton's **third law**, the Moon attracts the Earth with a force of the same magnitude 1.99×10^{20} N, opposite in direction (towards the Moon). The forces form an action–reaction pair: same magnitude, same line of action, opposite senses, acting on different bodies.

3. Centripetal force check.

For a circular orbit of radius r and period T, the required centripetal force on the Moon is

$$F_c = M_M r\omega^2 = M_M r\left(\frac{2\pi}{T}\right)^2.$$

Convert the period: $T = 27.3 \times 24 \times 3600 = 2.359 \times 10^{6}$ s.

$$F_c = (7.35 \times 10^{22})(3.84 \times 10^{8})\left(\frac{2\pi}{2.359 \times 10^{6}}\right)^2.$$

Compute the angular speed: $\omega = \dfrac{2\pi}{2.359 \times 10^{6}} = 2.663 \times 10^{-6}\ \text{rad s}^{-1}$, so $\omega^2 = 7.09 \times 10^{-12}\ \text{s}^{-2}$.

$$F_c = (7.35 \times 10^{22})(3.84 \times 10^{8})(7.09 \times 10^{-12}) = 2.00 \times 10^{20}\ \text{N}.$$

This equals the gravitational force found in part 1 (to within rounding). Hence gravity alone accounts for the Moon's orbital motion — historically this was Newton's great test of his law.

PROPERTY **Kepler's Laws of Planetary Motion**

1. First Law (Law of Ellipses): The orbit of a planet around the Sun is an ellipse with the Sun at one focus.
2. Second Law (Law of Equal Areas): A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time. This is a consequence of conservation of angular momentum.
3. Third Law (Harmonic Law): The square of the orbital period T of a planet is proportional to the cube of the semi-major axis a of its orbit:

   $$T^2 \propto a^3 \quad \text{or} \quad \frac{T^2}{a^3} = \text{constant for all planets orbiting the same star.}$$

   For circular orbits, $a = r$ (orbital radius).

EXPERIMENT **Verifying Kepler's Third Law with Solar System Data**

Using the mean orbital radii and periods of the planets (data from NASA), compute T^2/r^3 for each planet. You will find the ratio is constant (within observational uncertainty), confirming Kepler's third law. This law was later derived by Newton from his law of gravitation, showing that the constant is $4\pi^2/(GM_{\text{Sun}})$.

WORKED EXAMPLE 9 **Applying Kepler's Third Law**

The Earth orbits the Sun at a mean distance $r_E = 1.50 \times 10^{11}$ m with period $T_E = 365.25$ days. Mars has an orbital period $T_M = 687$ days. Assuming circular orbits, find the mean orbital radius of Mars.

SOLUTION

Using Kepler's third law as a ratio (same central mass, the Sun):

$$\left(\frac{T_M}{T_E}\right)^2 = \left(\frac{r_M}{r_E}\right)^3 \Rightarrow r_M = r_E\left(\frac{T_M}{T_E}\right)^{2/3}.$$

$$\frac{T_M}{T_E} = \frac{687}{365.25} = 1.881, \quad (1.881)^{2/3} = 1.524.$$

$$r_M = 1.50 \times 10^{11} \times 1.524 = 2.29 \times 10^{11}\ \text{m}.$$

This matches the accepted value for Mars' semi-major axis.

8.2 Gravitational Field Strength and Variation of g

DEFINITION **Gravitational Field Strength**

The gravitational field strength $\vec{g}$ at a point in a gravitational field is the gravitational force per unit mass experienced by a small test mass placed at that point:

$$\vec{g} = \frac{\vec{F}}{m}, \qquad \text{unit: N kg}^{-1}\ (\text{equivalent to m s}^{-2}).$$

It is a vector quantity, directed towards the centre of the mass creating the field.

PROPERTY **Field Strength of a Spherical Mass**

For a point mass M or a spherically symmetric body of mass M, the gravitational field

strength at a distance r from its centre is

$$g = \frac{GM}{r^2} \quad (r \geq R),$$

where R is the radius of the body. At the surface ($r = R$), $g_0 = GM/R^2$. At altitude h, $r = R + h$ and $g = GM/(R + h)^2$.

KEY POINTS **Field Strength vs. Weight**

- **Mass** m is an intrinsic property of a body (invariant, unit kg).
- **Weight** $W = mg$ is the gravitational force on the body at a specific location (depends on local g, unit N).
- Field strength g is a property of the field at a point; weight is a property of the body in that field.

METHOD **Using $g = GM/r^2$ and relating field strength to weight**

1. The **gravitational field strength** at a point is the force per unit mass: $\vec{g} = \vec{F}/m$, unit N kg^{-1} (equivalent to m s^{-2}).
2. At distance r from the centre of a spherical planet of mass M:

   $$g = \frac{GM}{r^2}.$$

   At the surface, $r = R$ (planet's radius); at altitude h, $r = R + h$.
3. Weight of a body of mass m at that point: $W = mg$.
4. For ratio problems (comparing two planets or two altitudes), write

   $$\frac{g_1}{g_2} = \frac{M_1}{M_2}\left(\frac{r_2}{r_1}\right)^2$$

   — constants cancel, no need for G.
5. Reflex: mass m is invariant; weight W depends on the local g.

WORKED EXAMPLE 10 **Weight of a mountaineer on Mount Cameroon**

Take $M_E = 5.98 \times 10^{24}$ kg, mean Earth radius $R_E = 6.37 \times 10^6$ m. A climber of mass $m = 70.0$ kg stands at the summit of Mount Cameroon, altitude $h = 4.10 \times 10^3$ m.

1. Calculate g at the summit and at sea level.
2. Calculate the climber's weight at both places and comment.

SOLUTION

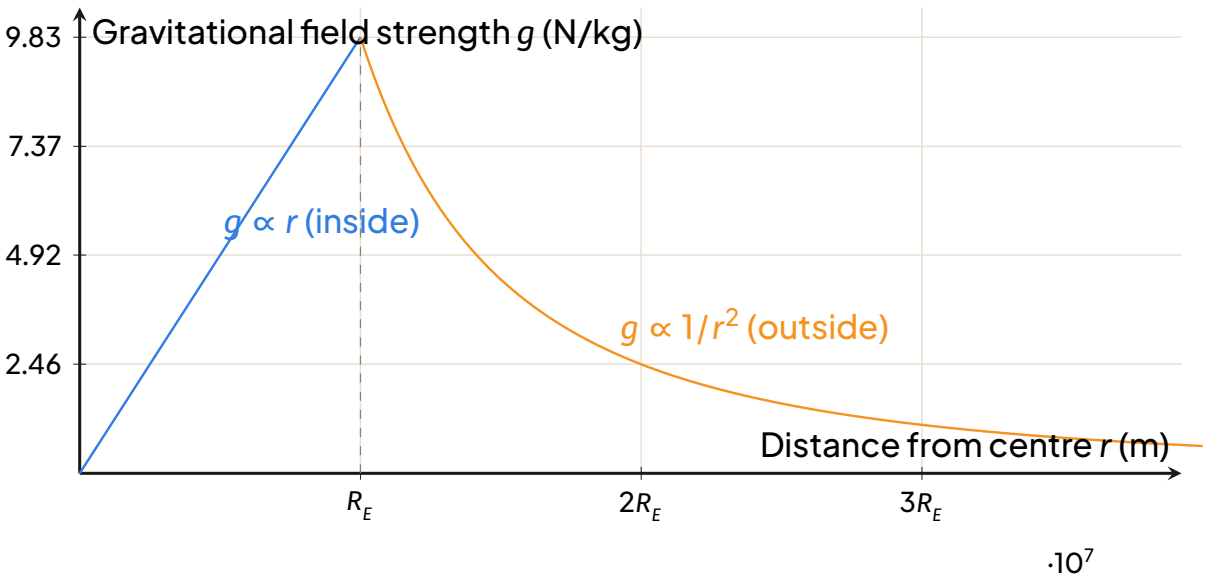
1. Field strengths.
At sea level, $r = R_E$:

$$g_0 = \frac{GM_E}{R_E^2} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{(6.37 \times 10^6)^2} = \frac{3.989 \times 10^{14}}{4.058 \times 10^{13}} = 9.83\ \text{N kg}^{-1}.$$

At the summit, $r = R_E + h = 6.37 \times 10^6 + 4.10 \times 10^3 = 6.3741 \times 10^6$ m:

$$g_h = \frac{3.989 \times 10^{14}}{(6.3741 \times 10^6)^2} = \frac{3.989 \times 10^{14}}{4.063 \times 10^{13}} = 9.82\ \text{N kg}^{-1}.$$

2. Weights.
At sea level: $W_0 = mg_0 = 70.0 \times 9.83 = 688$ N.
At the summit: $W_h = mg_h = 70.0 \times 9.82 = 687$ N.
Comment. The decrease is about 0.1 %: altitude variations of g on Earth are small because $h \ll R_E$. The climber's mass is unchanged (70.0 kg everywhere); only the weight varies. Note that the value $g = 9.81\ \text{m s}^{-2}$ used in class is a standard reference value; the local value depends on altitude and latitude.



Variation of gravitational field strength g with distance r from Earth's centre. Inside the Earth ($r < R_E$), g increases linearly with r assuming uniform density. Outside ($r > R_E$), g follows an inverse-square law.

METHOD **Choosing the correct formula for $g(r)$ inside and outside a uniform sphere**

1. Locate the point: is $r \geq R$ (outside) or $r \leq R$ (inside)? Sketch the situation.
2. Outside ($r \geq R$): the whole mass M acts as if at the centre (Shell Theorem):

$$g(r) = \frac{GM}{r^2} \quad \text{(inverse-square decrease).}$$

3. Inside ($r \leq R$), assuming uniform density ρ: only the mass within radius r contributes (shells outside exert zero net force). The enclosed mass is $M_{\text{in}} = \frac{4}{3}\pi r^3 \rho = M(r/R)^3$, giving

$$g(r) = \frac{GM_{\text{in}}}{r^2} = \frac{GM}{R^3}\, r = \frac{4}{3}\pi G \rho\, r \quad \text{(linear increase with } r\text{).}$$

4. Continuity check: both formulas give $g = GM/R^2$ at $r = R$. At the centre, $g = 0$.
5. Graph: g rises linearly from 0 at $r = 0$ to a maximum at $r = R$, then falls as $1/r^2$.

WORKED EXAMPLE 11 **Gravity in a deep mine**

Assume the Earth is a uniform sphere of mass $M = 5.98 \times 10^{24}$ kg and radius $R = 6.37 \times 10^{6}$ m, with surface field $g_0 = 9.83\ \text{N kg}^{-1}$.

1. Show that at depth d below the surface, $g = g_0\left(1 - \frac{d}{R}\right)$.
2. Calculate g at the bottom of a mine shaft $d = 3.0$ km deep.
3. Compare with the value at altitude $h = 3.0$ km above the surface. Which changes g more?

SOLUTION

1. Derivation.
At depth d, the distance from the centre is $r = R - d < R$, so we use the interior formula:

$$g(r) = \frac{GM}{R^3}\, r = \frac{GM}{R^3}(R - d).$$

Since $g_0 = \frac{GM}{R^2}$, we can write $\frac{GM}{R^3} = \frac{g_0}{R}$, hence

$$g = \frac{g_0}{R}(R - d) = g_0\left(1 - \frac{d}{R}\right). \quad \blacksquare$$

2. Value at $d = 3.0$ km.
$\frac{d}{R} = \frac{3.0 \times 10^3}{6.37 \times 10^6} = 4.71 \times 10^{-4}$.

$$g = 9.83\,(1 - 4.71 \times 10^{-4}) = 9.83 - 0.0046 = 9.825\ \mathrm{N\,kg^{-1}}.$$

3. Comparison with altitude $h = 3.0$ km.
Outside the Earth, $g = \frac{GM}{(R+h)^2} = g_0\left(\frac{R}{R+h}\right)^2$. Since $h \ll R$, use the binomial approximation $\left(1 + \frac{h}{R}\right)^{-2} \approx 1 - \frac{2h}{R}$:

$$g \approx g_0\left(1 - \frac{2h}{R}\right) = 9.83\,(1 - 9.42 \times 10^{-4}) = 9.821\ \mathrm{N\,kg^{-1}}.$$

Conclusion: ascending 3.0 km reduces g by about twice as much as descending 3.0 km. Going up, you move away from all the mass; going down, part of the mass is left above you and its pull cancels.

8.3 Gravitational Potential, Potential Energy, Escape and Orbital Speeds

DEFINITION **Gravitational Potential**

The gravitational potential V at a point in a gravitational field is the work done per unit mass by an external agent in bringing a test mass from infinity to that point, without accelera-

tion. For a point mass M or a spherically symmetric body:

$$V = -\frac{GM}{r} \quad (\mathrm{J\,kg^{-1}}),$$

with the zero of potential defined at infinity ($r \to \infty$). The potential is always negative for an attractive field.

DEFINITION **Gravitational Potential Energy**

The gravitational potential energy E_p of a system consisting of a mass m in the field of a mass M is the work done by an external agent to assemble the system from infinite separation:

$$E_p = mV = -\frac{GMm}{r} \quad (\mathrm{J}).$$

It is a property of the system (both masses), not of the test mass alone.

PROPERTY **Properties of V and E_p**

- Both are scalar quantities (easier to work with than vector forces).
- Both are negative for attractive fields, with zero at infinity.
- The potential V characterises the field itself (energy per unit mass); E_p characterises the body–field system.
- Difference in potential between two points A and B:

$$\Delta V = V_B - V_A = GM\left(\frac{1}{r_A} - \frac{1}{r_B}\right).$$

- Work done by an external agent moving mass m slowly from A to B:

$$W_{A\to B} = \Delta E_p = m\Delta V = GMm\left(\frac{1}{r_A} - \frac{1}{r_B}\right).$$

- For a body moving freely in the field, total mechanical energy is conserved:

$$E = \frac{1}{2}mv^2 - \frac{GMm}{r} = \text{constant}.$$

WARNING **Common Traps with Potential and Potential Energy**

- Sign convention: V and E_p are negative. Moving away from the planet (r increases) makes them less negative (i.e. they increase towards zero).
- Do not confuse V (field property, $\mathrm{J\,kg^{-1}}$) with E_p (system property, J).
- Do not use $E_p = mgh$ for large altitude changes — it assumes constant g and is only valid for $h \ll R$.
- Work done by gravity is $W_{\text{grav}} = -\Delta E_p$. Work done against gravity is $W_{\text{ext}} = +\Delta E_p$.

METHOD **Working with potential V and potential energy E_p**

1. Definitions (zero at infinity, both always negative for attractive fields):

$$V = -\frac{GM}{r} \quad (\mathrm{J\,kg^{-1}}), \qquad E_p = mV = -\frac{GMm}{r}.$$

2. Work done by an external agent moving mass m slowly from A to B:

$$W_{A\to B} = \Delta E_p = E_{p,B} - E_{p,A} = GMm\left(\frac{1}{r_A} - \frac{1}{r_B}\right).$$

3. Sign reflex: moving away from the planet ($r_B > r_A$) requires positive external work; E_p increases (becomes less negative) towards 0.
4. Energy conservation for a body moving freely in the field:

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = \text{constant}.$$

5. Trap: do not mix V (per unit mass, property of the field) with E_p (property of the body–planet system). Do not use $E_p = mgh$ over large altitude changes — it assumes constant g.

WORKED EXAMPLE 12 **Lifting a satellite into orbit**

A satellite of mass $m = 500$ kg is taken from the Earth's surface to a circular orbit of radius $r = 7.00 \times 10^6$ m. Take $M = 5.98 \times 10^{24}$ kg, $R = 6.37 \times 10^6$ m.

1. Calculate the gravitational potential at the surface and at the orbit.
2. Calculate the change in the satellite's gravitational potential energy.

3. Hence find the minimum work the launch system must supply against gravity.

SOLUTION

1. Potentials.
At the surface:

$$V_R = -\frac{GM}{R} = -\frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{6.37 \times 10^{6}} = -\frac{3.989 \times 10^{14}}{6.37 \times 10^{6}} = -6.26 \times 10^{7}\ \text{J kg}^{-1}.$$

At the orbit:

$$V_r = -\frac{GM}{r} = -\frac{3.989 \times 10^{14}}{7.00 \times 10^{6}} = -5.70 \times 10^{7}\ \text{J kg}^{-1}.$$

Both are negative; the potential is higher (less negative) in orbit.
2. Change in potential energy.

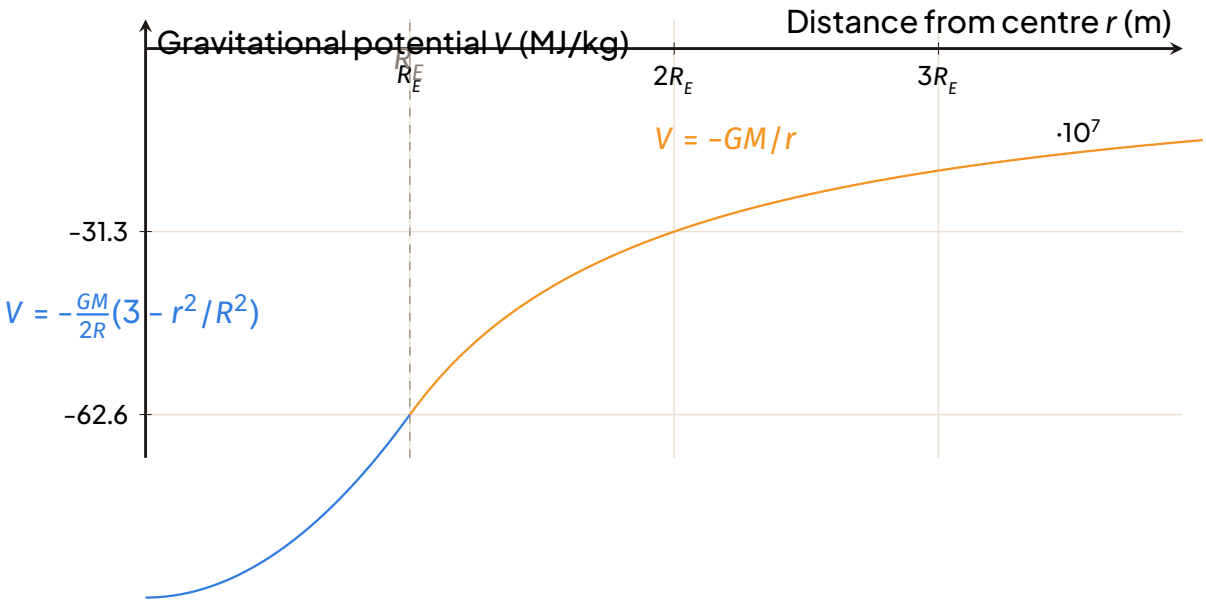
$$\Delta E_p = m(V_r - V_R) = 500 \times \left[(-5.70 \times 10^{7}) - (-6.26 \times 10^{7})\right] = 500 \times (5.6 \times 10^{6}).$$

$$\boxed{\Delta E_p = +2.8 \times 10^{9}\ \text{J}}$$

The positive sign confirms that energy must be supplied to move the satellite away from the Earth.
3. Minimum work against gravity.
If the satellite is moved slowly (no change in kinetic energy), the external work equals ΔE_p:

$$W_{\text{min}} = 2.8 \times 10^{9}\ \text{J}.$$

Remark: in reality the rocket must also supply the kinetic energy of the orbit, $\frac{1}{2}mv^2 = \frac{GMm}{2r} = \frac{(3.989 \times 10^{14})(500)}{2 \times 7.00 \times 10^{6}} = 1.42 \times 10^{10}$ J — actually larger than the potential-energy gain!
The total mechanical energy to be supplied is about 1.70×10^{10} J.



Gravitational potential V vs distance r for a uniform sphere. Inside ($r < R_E$), V is parabolic; outside, it follows $-1/r$. The potential is continuous and differentiable at $r = R_E$.

METHOD **Deriving and using v_{esc} and v_{orb}**

1. Escape velocity (from surface R): set total mechanical energy to zero (body just reaches infinity at rest):

$$\frac{1}{2}mv_{esc}^2 - \frac{GMm}{R} = 0 \Rightarrow v_{esc} = \sqrt{\frac{2GM}{R}} = \sqrt{2g_0R}.$$

2. Orbital speed for a circular orbit of radius r: equate gravity to centripetal force:

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow v_{orb} = \sqrt{\frac{GM}{r}}.$$

3. Memory hook: $v_{esc} = \sqrt{2}\, v_{orb}$ at the same radius.

4. Period: $T = \frac{2\pi r}{v_{orb}} = 2\pi\sqrt{\frac{r^3}{GM}}$ — this is Kepler's third law, $T^2 \propto r^3$.

5. Reflex: both speeds are independent of the satellite's mass m — it cancels.

WORKED EXAMPLE 13 **Escape from the Earth and from the Moon**

Take $M_E = 5.98\times10^{24}$ kg, $R_E = 6.37\times10^6$ m; for the Moon $M_M = 7.35\times10^{22}$ kg, $R_M = 1.74\times10^6$ m.

1. Derive the expression for escape velocity from the surface of a planet.
2. Calculate the escape velocities of the Earth and of the Moon.
3. A low Earth-orbit satellite circles at $r = 6.77 \times 10^6$ m. Find its orbital speed and period.

SOLUTION

1. Derivation.
Conservation of mechanical energy between the surface (speed v_{esc}) and infinity (speed 0, potential 0):

$$\underbrace{\frac{1}{2}mv_{esc}^2 - \frac{GMm}{R}}_{\text{surface}} = \underbrace{0 + 0}_{\text{infinity}} \Rightarrow v_{esc} = \sqrt{\frac{2GM}{R}}. \quad \blacksquare$$

2. Numerical values.
Earth:

$$v_{esc,E} = \sqrt{\frac{2(6.67 \times 10^{-11})(5.98 \times 10^{24})}{6.37 \times 10^6}} = \sqrt{\frac{7.977 \times 10^{14}}{6.37 \times 10^6}} = \sqrt{1.252 \times 10^8} = 1.12 \times 10^4\,\text{m s}^{-1} \approx 11.2\,\text{km s}^{-1}.$$

Moon:

$$v_{esc,M} = \sqrt{\frac{2(6.67 \times 10^{-11})(7.35 \times 10^{22})}{1.74 \times 10^6}} = \sqrt{\frac{9.806 \times 10^{12}}{1.74 \times 10^6}} = \sqrt{5.636 \times 10^6} = 2.37 \times 10^3\,\text{m s}^{-1} \approx 2.4\,\text{km s}^{-1}.$$

The Moon's low escape velocity explains why it cannot retain an atmosphere: fast gas molecules escape easily.
3. Orbital speed and period.

$$v_{orb} = \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{3.989 \times 10^{14}}{6.77 \times 10^6}} = \sqrt{5.892 \times 10^7} = 7.68 \times 10^3\,\text{m s}^{-1} \approx 7.7\,\text{km s}^{-1}.$$

$$T = \frac{2\pi r}{v_{orb}} = \frac{2\pi(6.77 \times 10^6)}{7.68 \times 10^3} = 5.54 \times 10^3\,\text{s} \approx 92\,\text{min}.$$

A low-orbit satellite circles the Earth in about an hour and a half — consistent with observations of the International Space Station.

8.4 Geostationary Satellites and Kepler's Third Law

DEFINITION **Geostationary Satellite**

A geostationary satellite orbits the Earth in the equatorial plane, in the same direction as the Earth's rotation, with an orbital period exactly equal to the Earth's sidereal rotation period (T = 23 h 56 min 4 s ≈ 8.616×10^4 s). It appears stationary relative to a fixed point on

the equator.

PROPERTY **Geostationary Orbit Parameters**

For a satellite of mass m in a circular geostationary orbit around Earth (mass M_E):

- Orbital radius: $r = \left(\frac{GM_E T^2}{4\pi^2}\right)^{1/3}$
- Altitude: $h = r - R_E$
- Orbital speed: $v = \frac{2\pi r}{T} = \sqrt{\frac{GM_E}{r}}$

The values are independent of the satellite's mass m.

METHOD **Solving geostationary / Kepler problems**

1. A **geostationary** satellite: circular orbit in the equatorial plane, same sense as the Earth's rotation, period $T = 24$ h (sidereal day, $\approx 8.616 \times 10^4$ s).
2. Find the radius from Kepler's third law:

$$\frac{T^2}{r^3} = \frac{4\pi^2}{GM} \Rightarrow r = \left(\frac{GMT^2}{4\pi^2}\right)^{1/3}.$$

3. Altitude: $h = r - R_E$. Speed: $v = 2\pi r/T$.
4. Comparing two satellites around the same planet: use the ratio

$$\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{r_1}{r_2}\right)^3$$ (no need for G or M).

5. Trap: $T^2/r^3 =$ constant holds only for satellites orbiting the same central mass.

WORKED EXAMPLE 14 **Placing a communications satellite over Cameroon**

A telecommunications company wants a geostationary satellite to serve Central Africa. Take $M_E = 5.98 \times 10^{24}$ kg, $R_E = 6.37 \times 10^6$ m, sidereal day $T = 8.616 \times 10^4$ s.

1. Calculate the radius and the altitude of the geostationary orbit.
2. Calculate the orbital speed of the satellite.

3. A second satellite has an orbital period of 12 h. Using Kepler's third law only, find the ratio of its orbital radius to that of the geostationary satellite.

SOLUTION

1. Radius and altitude.
From Kepler's third law:

$$r^3 = \frac{GMT^2}{4\pi^2} = \frac{(3.989 \times 10^{14})(8.616 \times 10^4)^2}{4\pi^2}.$$

Compute: $T^2 = 7.424 \times 10^9\ \text{s}^2$; numerator = $3.989 \times 10^{14} \times 7.424 \times 10^9 = 2.961 \times 10^{24}$; divide by $4\pi^2 = 39.48$:

$$r^3 = 7.50 \times 10^{22}\ \text{m}^3 \Rightarrow r = (7.50 \times 10^{22})^{1/3} = 4.22 \times 10^7\ \text{m}.$$

Altitude:

$$h = r - R_E = 4.22 \times 10^7 - 6.37 \times 10^6 = 3.58 \times 10^7\ \text{m} \approx 35\,800\ \text{km}.$$

2. Orbital speed.

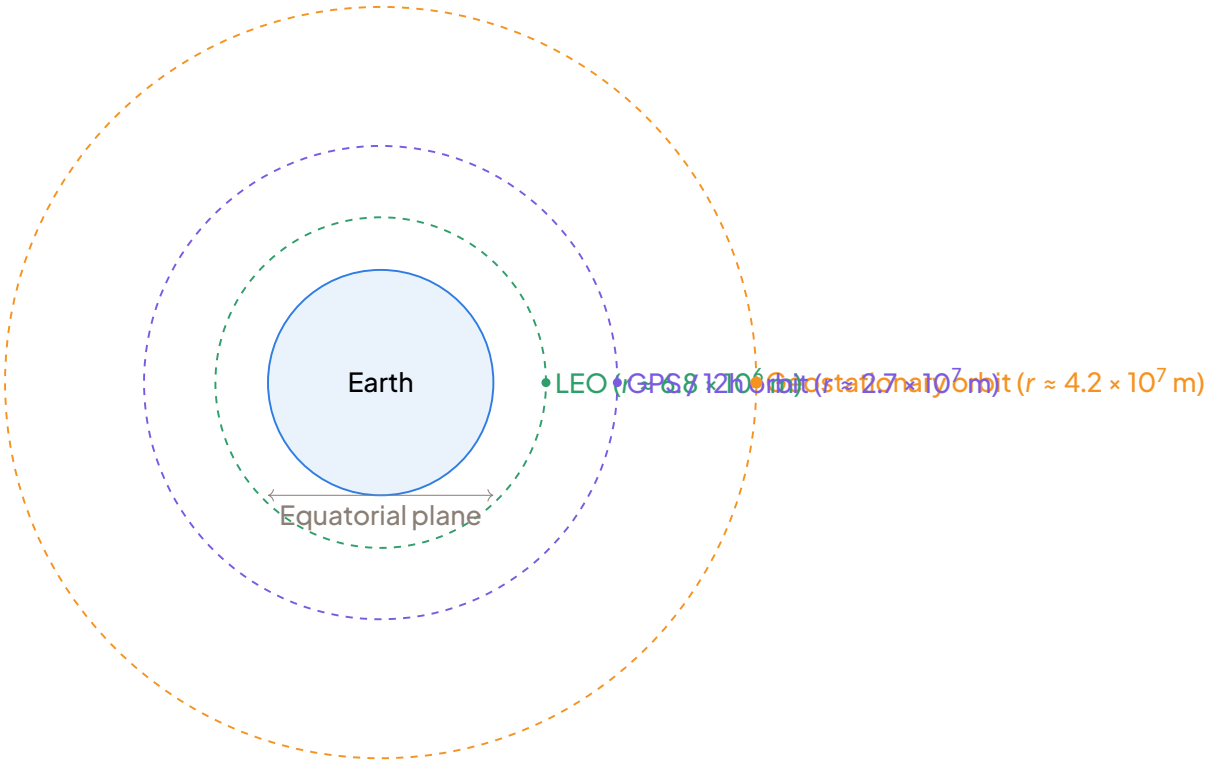
$$v = \frac{2\pi r}{T} = \frac{2\pi(4.22 \times 10^7)}{8.616 \times 10^4} = 3.08 \times 10^3\ \text{m s}^{-1} \approx 3.1\ \text{km s}^{-1}.$$

The satellite moves at about 3 km s^{-1} yet appears fixed in the sky, because the Earth rotates beneath it at the same angular rate — a dish antenna in Yaoundé, once pointed at it, never needs to move.

3. Ratio for the 12-hour satellite.
Both satellites orbit the Earth, so Kepler's third law applies as a ratio:

$$\frac{r_2}{r_1} = \left(\frac{T_2}{T_1}\right)^{2/3} = \left(\frac{12}{24}\right)^{2/3} = (0.5)^{2/3} = 0.63.$$

So $r_2 \approx 0.63 \times 4.22 \times 10^7 = 2.66 \times 10^7$ m. (This is roughly the altitude class of GPS-type navigation satellites.)



Comparison of satellite orbits around Earth (not to scale). Geostationary orbit is at ≈ 35 800 km altitude; GPS satellites orbit at ≈ 20 200 km (12 h period); Low Earth Orbit (LEO) satellites like the ISS orbit at ≈ 400 km (90 min period).

8.5 Motion in the Gravitational Field: Projectile Motion

DEFINITION **Projectile Motion in a Uniform Gravitational Field**

Near the Earth's surface, the gravitational field is approximately **uniform** (constant magnitude $g \approx 9.81\ \text{m s}^{-2}$, direction vertically downwards). A projectile is any body given an initial velocity and then moving under the influence of gravity alone (air resistance neglected). Its trajectory is a parabola.

PROPERTY **Equations of Projectile Motion**

Choose axes: Ox horizontal, Oy vertical (upwards positive); origin at launch point. Initial

velocity $\vec{v}_0$ at angle θ to the horizontal.

$$\begin{aligned}
\text{Components:} &\quad v_{0x} = v_0 \cos\theta, \quad v_{0y} = v_0 \sin\theta. \\
\text{Acceleration:} &\quad a_x = 0, \quad a_y = -g. \\
\text{Velocity:} &\quad v_x(t) = v_0 \cos\theta, \quad v_y(t) = v_0 \sin\theta - gt. \\
\text{Position:} &\quad x(t) = v_0 \cos\theta\, t, \quad y(t) = v_0 \sin\theta\, t - \tfrac{1}{2}gt^2. \\
\text{Trajectory (eliminate } t\text{):} &\quad y = x\tan\theta - \frac{g\,x^2}{2v_0^2\cos^2\theta} \quad \text{(parabola).}
\end{aligned}$$

For launch and landing at the same height ($y = 0$):

$$\begin{aligned}
\text{Time of flight:} &\quad T = \frac{2v_0\sin\theta}{g}. \\
\text{Range:} &\quad R = \frac{v_0^2\sin 2\theta}{g}. \\
\text{Maximum height:} &\quad H = \frac{v_0^2\sin^2\theta}{2g}.
\end{aligned}$$

Maximum range on level ground occurs at $\theta = 45^\circ$.

KEY POINTS **Key Reflexes for Projectile Problems**

- Horizontal velocity is constant ($a_x = 0$); vertical motion is uniformly accelerated ($a_y = -g$).
- Time t is the same for horizontal and vertical motions — it links the two.
- At the top of the trajectory, $v_y = 0$ but $v_x \neq 0$; acceleration is still g downwards.
- The trajectory is symmetric about the maximum height only if launch and landing heights are equal.
- Energy conservation: $\frac{1}{2}mv^2 + mgy = \frac{1}{2}mv_0^2$ (useful for speed at a given height).

METHOD **Solving projectile problems**

1. Choose axes: Ox horizontal, Oy vertical (upwards); origin at launch point. Near the Earth's surface g is constant: the field is **uniform**.
2. Resolve the initial velocity: $v_{0x} = v_0\cos\theta$, $v_{0y} = v_0\sin\theta$.

3. Write the equations of motion (only acceleration is $-g$ along y):

$$x(t) = v_0 \cos\theta\, t, \qquad y(t) = v_0 \sin\theta\, t - \tfrac{1}{2} g t^2.$$

4. Trajectory: eliminate t:

$$y = x \tan\theta - \frac{g\,x^2}{2v_0^2 \cos^2\theta} \quad \text{(parabola).}$$

5. Key results (level ground):

$$\text{range } R = \frac{v_0^2 \sin 2\theta}{g}, \qquad \text{max height } H = \frac{v_0^2 \sin^2\theta}{2g}, \qquad \text{time of flight } T = \frac{2v_0 \sin\theta}{g}.$$

6. Reflexes: horizontal velocity is constant; at the summit $v_y = 0$ but $v_x \neq 0$; maximum range on level ground occurs at $\theta = 45^\circ$.

WORKED EXAMPLE 15 **A football free kick**

During a match in Bafoussam, a player kicks a ball from ground level with speed $v_0 = 20.0\ \text{m s}^{-1}$ at $\theta = 30.0^\circ$ above the horizontal. Take $g = 9.81\ \text{m s}^{-2}$ and neglect air resistance.

1. Write the time equations $x(t)$ and $y(t)$.
2. Calculate the time of flight and the range.
3. Calculate the maximum height reached.
4. Determine the velocity (magnitude and direction) of the ball at the top of its flight.

SOLUTION

1. Time equations.
$v_{0x} = 20.0 \cos 30.0^\circ = 20.0 \times 0.866 = 17.3\ \text{m s}^{-1}$; $v_{0y} = 20.0 \sin 30.0^\circ = 10.0\ \text{m s}^{-1}$.

$$x(t) = 17.3\,t, \qquad y(t) = 10.0\,t - \tfrac{1}{2}(9.81)t^2 = 10.0\,t - 4.91\,t^2 \quad \text{(in metres, } t \text{ in s).}$$

2. Time of flight and range.
The ball lands when $y = 0$ (excluding $t = 0$):

$$t(10.0 - 4.91\,t) = 0 \;\Rightarrow\; T = \frac{10.0}{4.91} = 2.04\ \text{s}.$$

(Check with the formula: $T = \dfrac{2v_0 \sin\theta}{g} = \dfrac{2 \times 20.0 \times 0.500}{9.81} = 2.04\ \text{s}$. ✓)

$$R = v_{0x}\,T = 17.3 \times 2.04 = 35.3\ \text{m}.$$

(Check: $R = \dfrac{v_0^2 \sin 2\theta}{g} = \dfrac{400 \times \sin 60.0^\circ}{9.81} = \dfrac{400 \times 0.866}{9.81} = 35.3\,\text{m}$. ✓)

3. Maximum height.

At the summit $v_y = 0$: $v_{0y} - gt = 0 \Rightarrow t = \dfrac{10.0}{9.81} = 1.02\,\text{s}$ (half the flight time, as expected from symmetry).

$$H = 10.0(1.02) - 4.91(1.02)^2 = 10.2 - 5.11 = 5.10\,\text{m}.$$

(Check: $H = \dfrac{v_0^2 \sin^2\theta}{2g} = \dfrac{400 \times 0.250}{19.62} = 5.10\,\text{m}$. ✓)

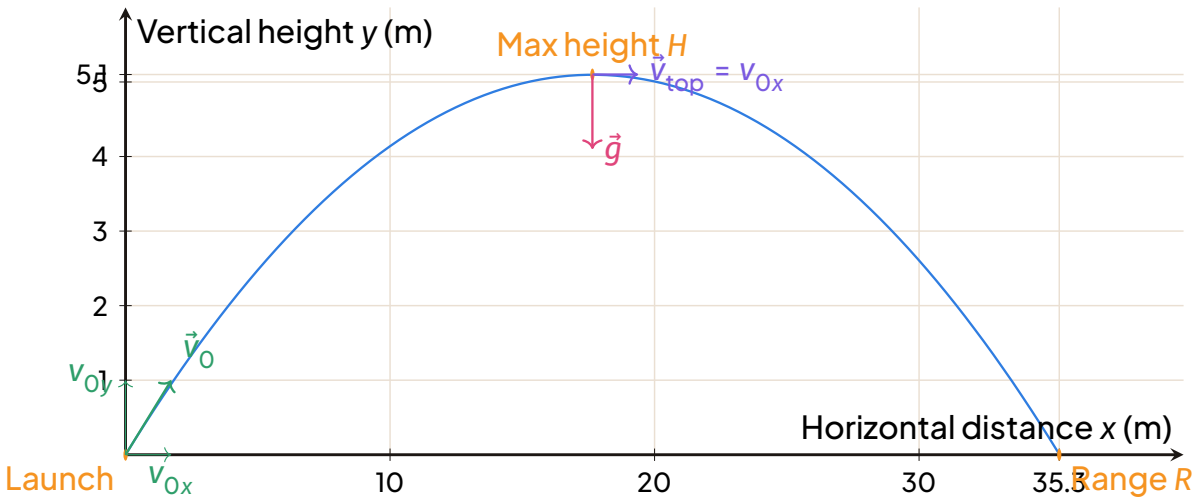
4. Velocity at the top.

At the summit the vertical component vanishes; only the (constant) horizontal component remains:

$$\vec{v}_{\text{top}} = v_{0x}\,\vec{i} = 17.3\,\text{m s}^{-1}, \quad \text{directed horizontally, in the sense of motion.}$$

WARNING **Common traps in projectile questions**

- The acceleration at the top is not zero: it is $g = 9.81\,\text{m s}^{-2}$ downwards throughout the flight. Only v_y is zero at the summit.
- Never mix components: time t is the same in $x(t)$ and $y(t)$ — it is the bridge between the two motions.
- The range formula $R = v_0^2 \sin 2\theta / g$ is valid only when launch and landing are at the same height.
- For launch from a height h_0 above ground, solve $y(t) = -h_0$ for t to find the time of flight.



Projectile trajectory for $v_0 = 20$ m/s, $\theta = 30^\circ$. The velocity vector is tangent to the path; at the top it is purely horizontal. Acceleration $\vec{g}$ is constant and vertically downward everywhere.

DID YOU KNOW? **Did you know?**

- The concept of a geostationary orbit was first proposed by Arthur C. Clarke in 1945 in a paper titled "Extra-Terrestrial Relays". Today, the geostationary orbit is often called the "Clarke Orbit" in his honour.
- Cameroon (like all equatorial countries) benefits greatly from geostationary satellites for telecommunications, weather monitoring, and broadcasting. A dish pointed at a geostationary satellite in Yaoundé never needs to track it.
- The escape velocity from the Solar System at Earth's orbit is about 42 km/s, but Earth's orbital speed is 30 km/s. Space probes use "gravity assists" (slingshot manoeuvres) around planets to gain the extra speed needed to leave the Solar System.
- The value $g = 9.81\,\text{m/s}^2$ is a standard average. At the poles, $g \approx 9.83\,\text{m/s}^2$; at the equator, $g \approx 9.78\,\text{m/s}^2$ due to Earth's rotation and equatorial bulge.

EXERCISE 6 **Practice Problems**

1. Two identical spheres of mass $m = 50$ kg each are placed with their centres 1.0 m apart. Calculate the gravitational force between them. Compare this force with the weight of one sphere.
2. A planet has twice the mass of Earth and twice the radius of Earth. What is the gravitational field strength at its surface in terms of g_E?

3. A satellite of mass 200 kg is in a circular orbit of radius 8.0×10^6 m around Earth. Calculate its (a) orbital speed, (b) period, (c) kinetic energy, (d) potential energy, (e) total mechanical energy.

4. Derive an expression for the minimum energy required to move a satellite of mass m from a circular orbit of radius r_1 to a higher circular orbit of radius r_2 ($r_2 > r_1$).

5. A projectile is launched from a cliff of height $h = 50$ m with speed $v_0 = 30$ m/s at $\theta = 40^\circ$ above horizontal. Find (a) the time of flight, (b) the horizontal range, (c) the maximum height above the cliff, (d) the speed just before impact.

6. A geostationary satellite orbits at $r = 4.22 \times 10^7$ m. A spy satellite orbits at $r = 7.00 \times 10^6$ m. How many orbits does the spy satellite complete in one day?

SOLUTION **Exercise 1**

$F = \frac{Gm^2}{r^2} = \frac{(6.67\times10^{-11})(50)^2}{1^2} = 1.67 \times 10^{-7}$ N. Weight $W = mg = 50 \times 9.81 = 490$ N. The gravitational force is $\approx 3.4 \times 10^{-10}$ times the weight — utterly negligible.

SOLUTION **Exercise 2**

$g_p = \frac{G(2M_E)}{(2R_E)^2} = \frac{2}{4}\frac{GM_E}{R_E^2} = \frac{1}{2}g_E$.

SOLUTION **Exercise 3**

(a) $v = \sqrt{GM/r} = \sqrt{3.989 \times 10^{14}/8.0 \times 10^6} = 7.06\times10^3$ m/s. (b) $T = 2\pi r/v = 7.12\times10^3$ s ≈ 119 min. (c) $K = \frac{1}{2}mv^2 = 4.98\times10^9$ J. (d) $U = -GMm/r = -9.97\times10^9$ J. (e) $E = K+U = -4.98\times10^9$ J $= -K = U/2$ (virial theorem for circular orbits).

SOLUTION **Exercise 4**

Total energy in circular orbit: $E = -\frac{GMm}{2r}$. Energy required: $\Delta E = E_2 - E_1 = \frac{GMm}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$. This is positive since $r_2 > r_1$.

SOLUTION **Exercise 5**

(a) Solve $y(t) = -50 = 30\sin 40^\circ\, t - 4.905t^2 \Rightarrow 4.905t^2 - 19.28t - 50 = 0$. Positive root: $t = 5.96\ \text{s}$. (b) $R = v_{0x}t = 30\cos 40^\circ \times 5.96 = 137\ \text{m}$. (c) $H_{\max} = \frac{(30\sin 40^\circ)^2}{2g} = 18.9\ \text{m}$ above cliff. (d) $v_y = v_{0y} - gt = 19.28 - 9.81 \times 5.96 = -39.2\ \text{m/s}$; $v_x = 22.98\ \text{m/s}$; $v = \sqrt{v_x^2 + v_y^2} = 45.5\ \text{m/s}$.

SOLUTION **Exercise 6**

$T_{\text{geo}} = 24\ \text{h}$. $T_{\text{spy}} = 2\pi\sqrt{r^3/GM} = 2\pi\sqrt{(7.00\times10^6)^3/3.989\times10^{14}} = 5.54\times10^3\ \text{s} = 1.54\ \text{h}$. Orbits per day $= 24/1.54 = 15.6$ orbits.

9 EXERCISES

9.1 I check my knowledge

EXERCISE 7 **MCQ: Newton's law of gravitation**

Two point masses m_1 and m_2 are separated by a distance r. The gravitational force between them is F. If the distance is doubled and each mass is tripled, the new force is:

1. $\frac{3F}{4}$
2. $\frac{9F}{4}$
3. $\frac{9F}{2}$
4. $9F$

EXERCISE 8 **True or false?**

State whether each of the following statements is true or false, and justify your answer in one or two sentences.

1. The gravitational field strength at the centre of the Earth is zero.
2. The gravitational potential at a point near the Earth is a positive quantity.
3. A geostationary satellite can be placed in orbit directly above Douala.
4. The escape velocity from a planet depends on the mass of the object being launched.

EXERCISE 9 **Definitions**

Define each of the following terms precisely, giving the defining formula and SI unit where appropriate:

1. gravitational field strength at a point;
2. gravitational potential at a point;
3. escape velocity;
4. geostationary satellite.

EXERCISE 10 **Direct calculation: force between lead spheres**

Two small lead spheres of masses $m_1 = 50\,\text{kg}$ and $m_2 = 80\,\text{kg}$ have their centres 0.50 m apart. Take $G = 6.67e-11\,\text{N m}^2\,\text{kg}^{-2}$.

1. Calculate the gravitational force of attraction between the spheres.
2. Compare this force with the weight of a sheet of paper of mass 1.0 g (take $g = 9.81\,\text{m s}^{-2}$).
3. Hence explain why gravitational forces between everyday objects are never felt in daily life.

9.2 I practise

EXERCISE 11 **Variation of g with position**

Take the mass of the Earth $M_E = 5.98e24\,\text{kg}$, its radius $R_E = 6.38e6\,\text{m}$ and $G = 6.67e-11\,\text{N m}^2\,\text{kg}^{-2}$. Assume the Earth has uniform density.

1. Calculate the gravitational field strength g_0 at the Earth's surface.
2. Calculate the value of g at an altitude $h = 2R_E$ above the surface.
3. Calculate the value of g at a depth $d = R_E/2$ below the surface.
4. Sketch the graph of $g(r)$ for r ranging from 0 to $3R_E$, marking the values found above.

EXERCISE 12 **A strange planet**

A planet has twice the mass of the Earth and half its radius.

1. Show that the gravitational field strength at its surface is $g_P = 8\,g_0$, where g_0 is the field strength at the Earth's surface.
2. Show that the escape velocity from this planet is $v_P = 2\,v_E$, where v_E is the escape velocity from the Earth.
3. Taking $g_0 = 9.81\,\text{m s}^{-2}$ and $v_E = 11.2\,\text{km s}^{-1}$, give the numerical values of g_P and v_P.

EXERCISE 13 **A low-orbit satellite**

A satellite is in a circular orbit around the Earth with a period of 90 minutes. Take $M_E =$

$5.98e24$ kg and $G = 6.67e-11\ \mathrm{N\,m^2\,kg^{-2}}$.

1. Using Kepler's third law, calculate the radius of the orbit.
2. Calculate the orbital speed of the satellite.
3. State, with justification, whether this satellite can be geostationary.

EXERCISE 14 **Projectile from a cliff in Limbe**

A stone is thrown horizontally with a speed of $15\ \mathrm{m\,s^{-1}}$ from the top of a cliff 45 m high overlooking the sea at Limbe. Air resistance is negligible; take $g = 9.81\ \mathrm{m\,s^{-2}}$.

1. Write down the equations of motion $x(t)$ and $y(t)$ of the stone, choosing axes clearly.
2. Calculate the time taken for the stone to reach the sea.
3. Calculate the horizontal distance from the foot of the cliff to the point of impact.
4. Determine the velocity (magnitude and direction below the horizontal) of the stone at impact.

9.3 I prepare for the GCE

EXERCISE 15 **Geostationary satellites (GCE style)**

1. State Kepler's three laws of planetary motion. (3 marks)
2. Define a geostationary satellite and state two of its uses for a country such as Cameroon. (3 marks)
3. Starting from Newton's law of gravitation and the condition for uniform circular motion, show that for a satellite of mass m in a circular orbit of radius r around the Earth (mass M_E),
$$T^2 = \frac{4\pi^2 r^3}{GM_E}.$$
(4 marks)
4. Hence calculate the orbital radius, the altitude above the Earth's surface, and the orbital speed of a geostationary satellite. (Take $G = 6.67e-11\ \mathrm{N\,m^2\,kg^{-2}}$, $M_E = 5.98e24$ kg, $R_E = 6.38e6$ m, $T = 24$ h.) (6 marks)
5. Explain, with the aid of a diagram, why a network of only three geostationary satellites can provide communication coverage of (almost) the whole Earth, and state which

regions are poorly covered. (4 marks)

EXERCISE 16 **Energy of an orbiting satellite (GCE style)**

A satellite of mass $m = 100$ kg is in a circular orbit of radius $r_1 = 2R_E$ around the Earth.

1. Define gravitational potential and gravitational potential energy. (3 marks)
2. Show that the total mechanical energy of a satellite in a circular orbit of radius r is
$$E = -\frac{GM_E m}{2r}.$$
(4 marks)
3. Explain the physical significance of the negative sign of this energy. (2 marks)
4. Calculate the work that must be supplied to transfer the satellite from the orbit of radius $r_1 = 2R_E$ to a higher circular orbit of radius $r_2 = 3R_E$. (Take $G = 6.67e-11\,\text{N m}^2\,\text{kg}^{-2}$, $M_E = 5.98e24$ kg, $R_E = 6.38e6$ m.) (5 marks)
5. The satellite is then allowed to fall back to Earth. Explain qualitatively what happens to its kinetic energy, potential energy and total mechanical energy if air resistance is taken into account. (3 marks)

EXERCISE 17 **Projectile motion and data analysis (GCE style)**

Part A. A football is kicked from level ground with an initial speed of $20\ \text{m s}^{-1}$ at an angle of 40° above the horizontal. Air resistance is negligible; take $g = 9.81\ \text{m s}^{-2}$.

1. Resolve the initial velocity into horizontal and vertical components. (2 marks)
2. Calculate the time of flight of the ball. (3 marks)
3. Calculate the maximum height reached. (3 marks)
4. Calculate the horizontal range. (2 marks)
5. State the angle of projection that gives the maximum range for a given initial speed, and justify your answer using the range formula. (2 marks)

Part B. The table below gives the mean orbital radius r and the period T of four moons of Jupiter.

Moon	Io	Europa	Ganymede	Callisto
r ($\times 10^8$ m)	4.22	6.71	10.7	18.8
T (days)	1.77	3.55	7.15	16.7

1. Copy and complete the table by computing T^2 (in s^2) and r^3 (in m^3). (3 marks)
2. Plot the graph of T^2 against r^3 and comment on its shape. (3 marks)
3. Use the gradient of the graph, together with Kepler's third law, to deduce the mass of Jupiter. (Take $G = 6.67e-11\ N\,m^2\,kg^{-2}$.) (4 marks)

10 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ: Newton's law of gravitation**

The initial force is $F = \dfrac{Gm_1m_2}{r^2}$. After the changes, the masses become $3m_1$ and $3m_2$ and the separation becomes $2r$:

$$F' = \frac{G(3m_1)(3m_2)}{(2r)^2} = \frac{9Gm_1m_2}{4r^2} = \frac{9}{4}F.$$

Correct answer: (b) $\dfrac{9F}{4}$.
Examiner's tip: the masses contribute a factor $3 \times 3 = 9$ (numerator), the distance a factor $2^2 = 4$ (denominator). A common error is to forget to square the distance factor, which would wrongly give $\frac{9F}{2}$.

SOLUTION **Exercise 2 — True or false?**

1. True. Inside a uniform spherical Earth, $g(r) = \dfrac{GM_E}{R_E^3}r$; at the centre $r = 0$, so $g = 0$. (Equivalently, by symmetry the pulls from all directions cancel.)
2. False. Gravitational potential is defined by $V = -\dfrac{GM}{r}$, with the zero chosen at infinity. Since the field is attractive, V is negative at every finite distance.
3. False. A geostationary satellite must orbit in the equatorial plane (above latitude 0°). Douala lies at about 4° N, so no satellite can remain fixed above it.
4. False. From $\frac{1}{2}mv_e^2 = \dfrac{GMm}{R}$, the test mass m cancels: $v_e = \sqrt{2GM/R}$ depends only on the planet's mass and radius, not on the mass of the launched object.

SOLUTION **Exercise 3 — Definitions**

1. Gravitational field strength at a point is the gravitational force per unit mass acting on a small test mass placed at that point:

   $$g = \frac{F}{m} \qquad \text{SI unit: N kg}^{-1} \text{ (equivalent to m s}^{-2}\text{).}$$

   It is a vector, directed towards the attracting mass.
2. Gravitational potential at a point is the work done per unit mass by an external agent

in bringing a small test mass from infinity to that point (zero of potential at infinity):

$$V = -\frac{GM}{r} \qquad \text{SI unit: J kg}^{-1}.$$

It is a scalar, always negative.

3. Escape velocity is the minimum speed with which an object must be launched from the surface of a planet so that it reaches infinity with zero residual speed:

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \qquad \text{SI unit: m s}^{-1}.$$

4. Geostationary satellite: a satellite in a circular orbit in the Earth's equatorial plane, with a period equal to the Earth's rotation period (T = 24 h) and revolving in the same sense as the Earth, so that it appears permanently fixed above one point of the equator (altitude ≈ 3.6e4 km).

SOLUTION **Exercise 4 — Direct calculation: force between lead spheres**

1. Applying Newton's law:

$$F = \frac{Gm_1m_2}{r^2} = \frac{(6.67 \times 10^{-11})(50)(80)}{(0.50)^2} = \frac{2.668 \times 10^{-7}}{0.25} \approx 1.07e-6\ \text{N}.$$

2. Weight of the sheet of paper:

$$W = mg = (1.0 \times 10^{-3})(9.81) = 9.81e-3\ \text{N}.$$

Ratio: $\frac{W}{F} = \frac{9.81 \times 10^{-3}}{1.07 \times 10^{-6}} \approx 9.2{\times}10^3$. The force between the spheres is about ten thousand times smaller than the weight of a single sheet of paper.

3. Because G is extremely small (~ 10^{-11} in SI units), gravitational forces between ordinary objects (kilogram masses at metre distances) are of order 10^{-7} N or less — far below friction and other contact forces, and undetectable by our senses. Gravitation only becomes appreciable when at least one of the masses is astronomical (e.g. the Earth).

SOLUTION **Exercise 5 — Variation of g with position**

1. At the surface:

$$g_0 = \frac{GM_E}{R_E^2} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{(6.38 \times 10^6)^2} = \frac{3.989 \times 10^{14}}{4.070 \times 10^{13}} \approx 9.80\ \text{N kg}^{-1}.$$

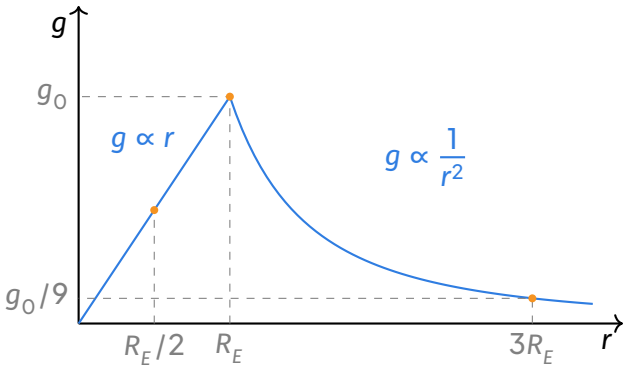
2. At altitude $h = 2R_E$, the distance from the centre is $r = R_E + h = 3R_E$:

$$g = \frac{GM_E}{(3R_E)^2} = \frac{g_0}{9} \approx \frac{9.80}{9} \approx 1.09\,\text{N kg}^{-1}.$$

3. Inside a uniform Earth, $g(r) = \frac{GM_E}{R_E^3} r = g_0 \frac{r}{R_E}$. At depth $d = R_E/2$, $r = R_E - d = R_E/2$:

$$g = g_0 \times \frac{1}{2} \approx 4.90\,\text{N kg}^{-1}.$$

4. Graph: g rises linearly from 0 at $r = 0$ to $g_0 \approx 9.80$ at $r = R_E$, then falls as $1/r^2$, passing through $g_0/9 \approx 1.09$ at $r = 3R_E$.



Variation of g with distance r from the centre of a uniform Earth.

SOLUTION **Exercise 6 — A strange planet**

1. Surface field strength:

$$g_P = \frac{GM_P}{R_P^2} = \frac{G(2M_E)}{(R_E/2)^2} = \frac{2GM_E}{R_E^2/4} = 8\frac{GM_E}{R_E^2} = 8g_0. \quad \blacksquare$$

2. Escape velocity:

$$v_P = \sqrt{\frac{2GM_P}{R_P}} = \sqrt{\frac{2G(2M_E)}{R_E/2}} = \sqrt{4\frac{2GM_E}{R_E}} = 2\sqrt{\frac{2GM_E}{R_E}} = 2v_E. \quad \blacksquare$$

3. Numerical values:

$$g_P = 8 \times 9.81 = 78.5\,\text{m s}^{-2}, \qquad v_P = 2 \times 11.2 = 22.4\,\text{km s}^{-1}.$$

Remark: halving the radius increases both g and v_e because they grow as the radius shrinks ($g \propto 1/R^2$, $v_e \propto 1/\sqrt{R}$).

SOLUTION **Exercise 7 — A low-orbit satellite**

1. Kepler's third law: $T^2 = \dfrac{4\pi^2 r^3}{GM_E}$, so $r = \left(\dfrac{GM_E T^2}{4\pi^2}\right)^{1/3}$. With $T = 90\ \text{min} = 5400\ \text{s}$:

$$r^3 = \frac{(6.67\times10^{-11})(5.98\times10^{24})(5400)^2}{4\pi^2} = \frac{(3.989\times10^{14})(2.916\times10^{7})}{39.48} \approx 2.946\times10^{20}\ \text{m}^3,$$

$$r \approx (2.946\times10^{20})^{1/3} \approx 6.66e6\ \text{m} \quad (\approx 6\,660\ \text{km}).$$

2. Orbital speed:

$$v = \frac{2\pi r}{T} = \frac{2\pi(6.66\times10^{6})}{5400} \approx 7.75e3\ \text{m s}^{-1} \approx 7.7\ \text{km s}^{-1}.$$

(Check: $v = \sqrt{GM_E/r} = \sqrt{3.989\times10^{14}/6.66\times10^{6}} \approx 7.74e3\ \text{m s}^{-1}$.)

3. No. A geostationary satellite must have $T = 24\ \text{h}$ and orbit in the equatorial plane at $r \approx 4.22e7\ \text{m}$. This satellite has $T = 90\ \text{min} \neq 24\ \text{h}$ (and its altitude is only $\approx 280\ \text{km}$), so it cannot be geostationary.

SOLUTION **Exercise 8 — Projectile from a cliff in Limbe**

Choose the origin O at the top of the cliff, x-axis horizontal in the direction of the throw, y-axis vertically downwards (so that g is positive). Initial conditions: $v_{0x} = 15\ \text{m s}^{-1}$, $v_{0y} = 0$.

1. Equations of motion:

$$x(t) = 15\,t, \qquad y(t) = \tfrac{1}{2}g\,t^2 = 4.905\,t^2 \quad (\text{SI units}).$$

2. The stone reaches the sea when $y = 45\ \text{m}$:

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times45}{9.81}} = \sqrt{9.174} \approx 3.03\ \text{s}.$$

3. Horizontal distance:

$$x = v_{0x}\,t = 15\times3.03 \approx 45.4\ \text{m}.$$

4. Velocity components at impact:

$$v_x = 15\ \text{m s}^{-1}, \qquad v_y = gt = 9.81\times3.03 \approx 29.7\ \text{m s}^{-1}.$$

Magnitude:

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{15^2 + 29.7^2} = \sqrt{225 + 882} \approx 33.3\ \text{m s}^{-1}.$$

Direction below the horizontal:

$$\tan\theta = \frac{v_y}{v_x} = \frac{29.7}{15} = 1.98 \quad \Longrightarrow \quad \theta \approx 63.2^\circ.$$

SOLUTION **Exercise 9 — Geostationary satellites (GCE style)**

1. Kepler's laws (3 marks — 1 per law):
   - 1st law (orbits): each planet moves on an ellipse with the Sun at one focus.
   - 2nd law (areas): the line joining the planet to the Sun sweeps out equal areas in equal times.
   - 3rd law (periods): the ratio $\frac{T^2}{r^3}$ is the same constant for all planets orbiting the Sun.
2. Definition (1 mark): a geostationary satellite is a satellite in a circular equatorial orbit whose period equals the Earth's rotation period (24 h) and which revolves in the same sense, so it appears fixed above one point of the equator. Uses (any 2, 1 mark each): television and radio broadcasting; telephone and internet relay; weather observation; GPS-type positioning support; monitoring of crops and forests.
3. Proof (4 marks). The gravitational attraction supplies the centripetal force:

$$\frac{GM_E m}{r^2} = mr\omega^2 \quad \text{(1 mark)}, \qquad \omega = \frac{2\pi}{T} \quad \text{(1 mark)}.$$

   Substituting:

$$\frac{GM_E}{r^2} = r\,\frac{4\pi^2}{T^2} \quad \Longrightarrow \quad T^2 = \frac{4\pi^2 r^3}{GM_E} \quad \text{(2 marks)}. \qquad \blacksquare$$

4. With $T = 24\ \text{h} = 86400\ \text{s}$:

$$r^3 = \frac{GM_E T^2}{4\pi^2} = \frac{(6.67\times10^{-11})(5.98\times10^{24})(86400)^2}{4\pi^2} = \frac{(3.989\times10^{14})(7.465\times10^{9})}{39.48} \approx 7.54\times10^{22}\ \text{m}^3,$$

$$r \approx 4.22e7\ \text{m} \approx 4.22\times10^{4}\ \text{km} \quad \text{(2 marks)}.$$

   Altitude: $h = r - R_E = 4.22\times10^{7} - 6.38\times10^{6} \approx 3.59e7\ \text{m} \approx 3.6\times10^{4}\ \text{km}$ (2 marks).

   Speed: $v = \frac{2\pi r}{T} = \frac{2\pi(4.22\times10^{7})}{86400} \approx 3.07e3\ \text{m s}^{-1} \approx 3.1\ \text{km s}^{-1}$ (2 marks).
5. Each geostationary satellite "sees" a disc covering roughly one third of the globe (about 120° of longitude). Three satellites spaced 120° apart in longitude therefore cover the whole equatorial and temperate belt (diagram: circle for the Earth, three

satellites at 120°, tangent rays from each satellite to the Earth's surface). The polar regions (latitudes above about 80°) are poorly covered because the satellites lie near the horizon there. (4 marks: diagram 1, idea of 120° spacing 1, coverage of the globe 1, polar regions 1.)

SOLUTION **Exercise 10 — Energy of an orbiting satellite (GCE style)**

1. Gravitational potential at a point: work done per unit mass in bringing a small test mass from infinity to that point; $V = -\dfrac{GM}{r}$ (J kg^{-1}). Gravitational potential energy of a mass m at that point: $E_p = mV = -\dfrac{GMm}{r}$ (J) — the work needed to bring m from infinity to the point. (3 marks)

2. Proof (4 marks). For a circular orbit, gravity provides the centripetal force:

$$\frac{GM_Em}{r^2} = \frac{mv^2}{r} \quad \Longrightarrow \quad E_k = \tfrac{1}{2}mv^2 = \frac{GM_Em}{2r} \quad \text{(2 marks).}$$

Total energy:

$$E = E_k + E_p = \frac{GM_Em}{2r} - \frac{GM_Em}{r} = -\frac{GM_Em}{2r} \quad \text{(2 marks).} \qquad \blacksquare$$

3. The negative sign means the satellite is in a bound state: its total energy is below the zero level (at infinity), so energy must be supplied to it ($\Delta E = +\frac{GM_Em}{2r}$) to free it from the Earth's attraction. (2 marks)

4. Work supplied = increase of total energy:

$$W = E_2 - E_1 = -\frac{GM_Em}{2r_2} + \frac{GM_Em}{2r_1} = \frac{GM_Em}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right) = \frac{GM_Em}{2R_E}\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{GM_Em}{12R_E}.$$

Numerically:

$$W = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(100)}{12(6.38 \times 10^{6})} = \frac{3.989 \times 10^{16}}{7.656 \times 10^{7}} \approx 5.2e8 \text{ J}.$$

(5 marks: expression of ΔE 2, simplification 1, numerical value with unit 2.)

5. As the satellite descends, its potential energy decreases (becomes more negative) and its kinetic energy would increase correspondingly; but air resistance (a non-conservative force) converts mechanical energy into heat, so the total mechanical energy decreases continuously. The satellite spirals inwards, heats up, and may burn up in the atmosphere. (3 marks: one per correct statement about E_k, E_p, E.)

SOLUTION **Exercise 11 — Projectile motion and data analysis (GCE style)**

Part A.

1. Components (2 marks):

$$v_{0x} = 20\cos 40^\circ = 15.3\ \text{m s}^{-1}, \qquad v_{0y} = 20\sin 40^\circ = 12.9\ \text{m s}^{-1}.$$

2. Time of flight — the ball lands when $y = v_{0y}t - \tfrac{1}{2}gt^2 = 0$, giving (3 marks):

$$T = \frac{2v_{0y}}{g} = \frac{2 \times 12.86}{9.81} \approx 2.62\ \text{s}.$$

3. Maximum height (3 marks):

$$H = \frac{v_{0y}^2}{2g} = \frac{(12.86)^2}{2 \times 9.81} = \frac{165.4}{19.62} \approx 8.43\ \text{m}.$$

4. Range (2 marks):

$$R = v_{0x}\,T = 15.32 \times 2.62 \approx 40.2\ \text{m}.$$

5. The range formula $R = \dfrac{v_0^2 \sin 2\theta}{g}$ is maximal when $\sin 2\theta = 1$, i.e. $2\theta = 90^\circ$, so $\theta = 45^\circ$. (2 marks)

Part B.

1. Converting T to seconds (1 day = 86400 s) and computing (values to 3 s.f.):

Moon	Io	Europa	Ganymede	Callisto
r^3 ($\times 10^{25}$ m^3)	7.51	30.2	122	664
T^2 ($\times 10^9$ s^2)	2.34	9.41	38.2	208

(3 marks: correct conversion 1, correct powers 2.)

2. The graph of T^2 against r^3 is a straight line through the origin (3 marks: axes labelled with units 1, correct points 1, straight line and comment 1). This confirms Kepler's third law: $T^2 \propto r^3$.

3. Kepler's third law gives $T^2 = \dfrac{4\pi^2}{GM_J} r^3$, so the gradient of the graph is $k = \dfrac{4\pi^2}{GM_J}$. From the table, using e.g. Callisto:

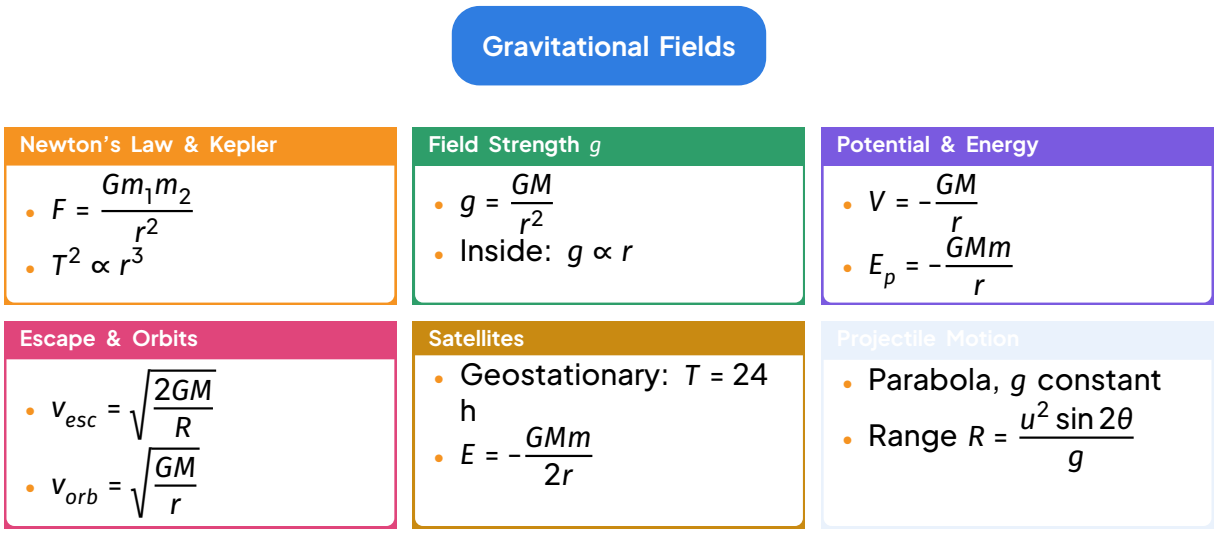
$$k = \frac{T^2}{r^3} = \frac{2.08 \times 10^{11}}{6.64 \times 10^{26}} \approx 3.13 \times 10^{-16}\ \text{s}^2\ \text{m}^{-3}.$$

Hence

$$M_J = \frac{4\pi^2}{Gk} = \frac{39.48}{(6.67 \times 10^{-11})(3.13 \times 10^{-16})} \approx 1.9e27\ \text{kg}.$$

(4 marks: linking gradient to $4\pi^2/GM_J$ 1, reading the gradient 1, correct algebra 1, value $\approx 1.9 \times 10^{27}$ kg 1.) This is in excellent agreement with the accepted mass of Jupiter.

11 SUMMARY SHEET



Mind map of the chapter: Gravitational Fields.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- **Gravitational field**: region of space around a mass in which another mass experiences an attractive force.
- **Gravitational field strength** g: force per unit mass placed at the point, $\vec{g} = \vec{F}/m$; unit N kg^{-1} (equivalent to m s^{-2}). It is a vector, directed towards the attracting mass.
- **Gravitational potential** V at a point: work done per unit mass by an external agent in bringing a small test mass from infinity to that point; unit J kg^{-1}. It is a scalar, always negative (zero at infinity).
- **Gravitational potential energy** E_p: work done to bring a mass m from infinity to the point; $E_p = mV$.
- **Escape velocity**: minimum launch speed for a body to leave a planet's attraction and reach infinity with zero residual speed.
- **Geostationary satellite**: satellite in a circular equatorial orbit whose period equals the Earth's rotation period (T = 24 h), so it appears fixed above one point of the equator.

Laws and formulas to know

- **Newton's law of gravitation** (inverse-square law):

$$F = \frac{Gm_1m_2}{r^2}, \qquad G = 6.67e-11\,\mathrm{N\,m^2\,kg^{-2}}.$$

  Attractive, acts along the line joining the centres; valid for point masses and spherical bodies (r measured centre to centre).

- **Kepler's laws**: (1) planets move on ellipses with the Sun at one focus; (2) the radius vector sweeps equal areas in equal times; (3) $T^2 \propto r^3$, i.e. $\frac{T^2}{r^3} = \frac{4\pi^2}{GM}$ for circular orbits.

- Field strength of a spherical mass M: $g = \frac{GM}{r^2}$ outside ($r \geq R$); at the surface $g_0 = \frac{GM}{R^2}$.

- **Variation of g**: at altitude h, $g_h = \frac{GM}{(R+h)^2} = g_0\frac{R^2}{(R+h)^2}$; at depth d (uniform Earth), $g_d = g_0\left(1-\frac{d}{R}\right)$, since only the inner sphere of radius $(R-d)$ attracts. Inside a uniform sphere, $g \propto r$; at the centre $g = 0$.

- Potential and energy:

$$V = -\frac{GM}{r}, \qquad E_p = -\frac{GMm}{r}, \qquad g = -\frac{dV}{dr}.$$

- Escape and orbital speeds:

$$v_{esc} = \sqrt{\frac{2GM}{R}} = \sqrt{2g_0R}, \qquad v_{orb} = \sqrt{\frac{GM}{r}} = \sqrt{\frac{g_0R^2}{r}}, \qquad T = 2\pi\sqrt{\frac{r^3}{GM}}.$$

- Energy of an orbiting satellite: $E_k = \frac{GMm}{2r}$, $E_p = -\frac{GMm}{r}$, total $E = -\frac{GMm}{2r}$ (bound, negative).

- Projectile motion (uniform g): horizontal $x = ut\cos\theta$; vertical $y = ut\sin\theta - \frac{1}{2}gt^2$; trajectory $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$ (parabola); time of flight $t_f = \frac{2u\sin\theta}{g}$; maximum height $H = \frac{u^2\sin^2\theta}{2g}$; range $R = \frac{u^2\sin 2\theta}{g}$, maximum at $\theta = 45^\circ$.

Methods

- Satellite problems: equate gravitational force to centripetal force, $\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2$, then extract v, ω or T.

- Energy problems: apply conservation of mechanical energy $E_k + E_p$ = constant between two points (e.g. surface and infinity for escape velocity).

- Projectiles: resolve the initial velocity into components; treat horizontal (uniform) and vertical (uniformly accelerated) motions independently; link them through the common time t.

- Always measure r from the centre of the Earth: $r = R + h$.

Common mistakes

- Using the altitude h instead of $r = R + h$ in $g = GM/r^2$ or v_{orb}.
- Forgetting the negative sign of V and E_p; potential increases (becomes less negative) with distance.
- Confusing $v_{esc} = \sqrt{2gR}$ with $v_{orb} = \sqrt{gR}$ at the surface: they differ by a factor $\sqrt{2}$.
- Writing $T^2 \propto r^2$ instead of $T^2 \propto r^3$ (Kepler's third law).
- Mixing units: convert km to m, hours to seconds, before substituting.
- Believing a satellite is "weightless" because there is no gravity: gravity supplies the centripetal force; the astronaut is in free fall.

GCE tips

- Quote Newton's law in words and symbols; state that G is the universal gravitational constant (do not confuse G with g).
- Show the derivation $mg = \frac{GMm}{R^2} \Rightarrow g = \frac{GM}{R^2}$: it links field strength to the inverse-square law and earns method marks.
- For geostationary satellites, state the three conditions: circular orbit, equatorial plane, period $T = 24$ h in the direction of the Earth's rotation.
- Carry units through every numerical answer and give sensible significant figures (usually 3 s.f.).
- Sketch the variation of g with r: linear rise from the centre to R, then inverse-square decrease for $r > R$.

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